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Crustacean disparity through the Phanerozoic: comparing morphological and stratigraphic data

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Crustaceans have been an important component of marine diversity and biomass since the earliest Phanerozoic. With a relatively well-documented fossil record, they provide an excellent subject for a continuous study of disparity (≈bodyplan variety) from the Cambrian to the Recent. A data base of 135 morphological characters forms the basis for cladistic and morphospace studies at the ordinal and sub-ordinal level. Gross cladistic topology is: (Eumalacostraca + Hoplocarida vs Maxillopoda) vs Phyllopoda (paraphyletic). Each of these groups is of approximately equal disparity, and occupies a distinct region of the morphospace plot. A few problematical fossils (e.g. *Waptia* and *Odaraia*) fall close to the base of the tree. Comparison of the cladogram with stratigraphic range data indicates the location of probable ghost lineages, and randomization procedures provide a statistical test of the goodness of fit of a given set of stratigraphic ranges to a given tree topology. Disparity indices are calculated at series and stage intervals. Observed range data indicate that Cambrian disparity was approximately one third its present level. The Earliest Ordovician saw a marked decrease, with an increase and subsequent plateau through rest of the period. Increases through the Silurian and Devonian corresponded to the radiation of branchiopods, cephalocarids, and latterly the Eumalacostraca and Hoplocarida. By the end of the Carboniferous, observed disparity had reached over four fifths of Recent levels, and the remaining history of the group saw a gradual but slightly irregular increase up until the end of the Tertiary. Indices of disparity incorporating ghost lineages exhibit less marked peaks and troughs, with fewer perturbations overall. Cladistically-implied disparity in the Lower Cambrian is estimated at three quarters of that in the Recent. Rarefaction is used to compare actual levels of disparity at each time interval with the mean for a similar number of taxa selected randomly from the list of all realized bodyplans. Most intervals preserved a range of forms more disparate than the mean of random samples drawn from the pool of all the taxa considered. From the Triassic to the Recent this difference was intermittently significant. Once occupied, extremes of morphospace tend not to fall vacant again.

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ADDITIONAL KEYWORDS:—morphospace – problematica – stratigraphic congruence – ghost ranges – Gap Excess Ratio – Cambrian explosion.

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INTRODUCTION

Recent studies of disparity (as distinct from diversity) have demonstrated how some large clades have diversified into morphospace, and thereby come to define new bodyplans. This has had considerable impact on our understanding of several evolutionary radiations, most notably the Cambrian. Some studies have compared the Cambrian fauna with the Recent (e.g. Wills, Briggs & Fortey, 1994, 1998; Wills, 1998b), while others have examined the fortunes of particular clades through a portion of the Phanerozoic (e.g. Foote, 1989, 1991a,b, 1992b, 1994). The present study traces the disparity of a highly diverse and important clade (the crustaceans) from the Cambrian to the Recent, making comparisons at series and stage intervals.

The crustaceans exhibit enormous plasticity of form, having evolved many distinctive bodyplans since their origins (Schram, 1982; Briggs, 1992). The segmented body and serially homologous appendages provide great scope for variety (as in arthropods more generally), although there appear to be limitations on what can and cannot be a functioning arrangement of parts. Homoplasy within the clade is high, and many characters have conflicting distributions (Schram, 1978, 1986). There may also be superficial similarities in overall form and function for distantly-related taxa in similar habitats (Schram, 1982). Crustaceans have a reasonably good fossil record throughout the Phanerozoic (Briggs, Weedon & Whyte, 1993; Whatley, Siveter & Boomer, 1993), making them well-suited to a longitudinal study of disparity. They also preserve a large number of morphological characters, permitting meaningful comparisons with the living fauna.

Direct comparisons of phylogenetic and stratigraphic data have underpinned a

number of important recent studies (Huelsenbeck, 1994; Benton, 1995). Several of these have assayed large numbers of cladograms statistically to address broad-scale questions (e.g. Has palaeontological knowledge improved with research time? Are phylogenies for some higher taxa more congruent with the fossil record than others? Is there a difference in 'goodness of fit' for trees based on morphology versus those derived from sequence data?). Similar approaches can be used to deduce the locations of probable ghost lineages for particular groups in detail. This may further inform our understanding of a clade's radiation and dispersal in morphospace through time.

PRINCIPAL ISSUES IN CRUSTACEAN EVOLUTION

Diagnosing the Crustacea as a group is problematical, but Schram (1986) identified a number of features typical of *most* taxa. Similarly, Wills *et al.* (1995, 1998) found the clade to be well supported, but observed that many characters reverse within the group. Where lateral eyes are present, these are usually free or pedunculate beneath the carapace. The second appendages are usually differentiated as sensory antennae, followed by gnathobasic mandibles. The outer rami of the trunk appendages are organized around a longitudinal rachis, and a pair of mobile appendages are usually present on the telson. All pass through a nauplius or egg nauplius stage in ontogeny. The cephalon of the adult comprises five post-acronal segments [fewer in some fossils that have hitherto been excluded from the subphylum for this reason (e.g. Müller & Walossek, 1986)]. In contrast, molecular work (e.g. Adoette & Philippe, 1993; Averof & Akam, 1995) raises the possibility of a paraphyletic Crustacea, with the Hexapoda derived from their midst (see also Carpenter, 1906; Sharov, 1966).

Relationships within the group have been discussed in detail elsewhere (Wills, 1998a). The following issues are significant in considerations of crustacean evolution and disparity:

(1) Gross topology. The first thorough, parsimony-based analysis of crustacean subclasses and orders (Schram, 1986) resolved the non-phyllocarid Malacostraca (Eumalacostraca and Hoplocarida) as sister group to a Maxillopoda/Phyllopoda dichotomy, with the remipedes basal to all other taxa. Brusca & Brusca (1990), by contrast, placed all the Malacostraca as sister-group to the Maxillopoda, succeeded by the Branchiopoda, Cephalocarida and Remipedia in pectinate succession. More recent, molecular studies (e.g. Abele *et al.*, 1992; Spears & Abele, 1998) even cast doubt on the monophyly of these component clades. Character conflict is such that any conceivable cladogram will necessitate accepting a large amount of homoplasy. Moreover, all published phylogenies imply extensive ghost ranges.

(2) Integrity of the Maxillopoda. The Branchiura, Cirripedia, Copepoda and Mystacocarida were united as the 'Copepodoidea' by Beklemischev (1952, 1969), and the 'Maxillopoda' by Dahl (1956, 1963) (see also Bowman & Abele, 1982; Boxshall & Huys, 1989; Grygier, 1983; Müller, 1985; Müller & Walossek, 1988; Newman, Zullo & Withers, 1969; Newman, 1983). The Ostracoda are usually considered members of this group (e.g. Siewing, 1960; Schram, 1986), as are the Tantulocarida (Grygier, 1987). Several recent morphological (e.g. Wilson, 1992) and molecular studies (e.g. Abele *et al.* 1992) have cast doubt on the monophyly of the group,

implying that the maxillopods have been tenuously united by a small number of plesiomorphic features.

(3) Status of the Phyllocarida and 'Phyllopoda'. Phyllocarids have traditionally been classified together with the Eumalacostraca and Hoplocarida in the Malacostraca (Claus, 1888; Metschnikov, 1968). Schram (1986) preferred to place them with the Cephalocarida and Branchiopoda in the class Phyllopoda. Few characters unite the phyllocarids themselves, particularly if one includes the canadaspidids (Briggs, 1978, 1992). The thorax and part of the abdomen are enclosed within a bivalved carapace. The thoracic limbs have lost all ambulatory function, and serve for respiration, feeding and brood protection (Dahl, 1976). Several primitive features are evident in their embryology (Manton, 1934), and in the long, tubular nature of the heart.

(4) Primitive groups and the 'urcrustacean'. Much recent carcinological theory has been underpinned by attempts to determine the most plesiomorphic crustacean bodyplan. Three extant groups have attracted most attention in this context. The Cephalocarida (Sanders, 1963; Hessler & Newman, 1975; Hessler, 1992) are elongate forms with phyllopodous appendages on the first nine of 19 trunk somites. The carapace-less Anostraca (Calman, 1909; Zimmer, 1926–27; Tiegs & Manton, 1958; Fryer, 1992) have 20 or more trunk somites, of which numbers 11 to 19 bear phyllopods. The more recently-discovered Remipedia (Yager, 1981; Schram, 1983; Schram & Emerson, 1986) have as many as 32 podous trunk somites behind a cephalon with several raptorial mouth parts. The ventrolaterally-directed trunk limbs are homonomous, biramous and oar-like. Common to all these taxa is a relatively elongate trunk, with little appendage differentiation. The remipedes lack all trunk tagmosis. Importantly, none of these groups are among the oldest supposed crustacean fossils (phyllocarids, ostracods and cirripedes), implying appreciable ghost ranges. All previous work on arthropod phylogeny by the author (e.g. Briggs, Fortey & Wills, 1992; Wills *et al.*, 1994, 1995, 1998) has placed the remipedes at the base of the crustacean clade. The extant Nectiopoda and fossil Enantiopoda are therefore used to root the analyses here (Schram, 1986). Several authors (Hessler & Newman, 1975; Schram, 1982; Fryer, 1992) removed the supposed apomorphies from the group that they believed most primitive in order to derive a hypothetical ancestor (urcrustacean). Only Fryer's model (Fryer, 1992) has a differentiated abdomen.

(5) The significance of problematic fossil taxa. A number of fossil taxa, particularly those from the Cambrian, defy easy assignment to any established crustacean subclass. Others lack the traditionally diagnostic cephalic segmentation for the entire group. Exquisitely preserved, phosphatized specimens from the Upper Cambrian Orsten fauna of Sweden have attracted considerable attention. Controversy centres around whether these fossils are simply larval 'flotsam' of uncertain affinities (Schram, 1986) or earliest representatives of the Anostraca (*Rehbachiella*: Walossek, 1993), Copepoda (*Skara*: Müller, 1985; Müller & Walossek, 1988), Thecostraca (*Bredocaris*: Müller & Walossek, 1988), and the stem lineage leading to all the crustaceans (*Martinssonia*: Müller & Walossek, 1986). Other anomalous taxa are documented from the Burgess Shale. *Waptia* is a slender, bivalved form, with possible affinities to the branchiopods (Briggs, 1983; but see Briggs *et al.*, 1992). *Odaraia* (Briggs, 1981) is bivalved, with 25–30 trunk somites each bearing biramous appendages. It shares derived characters with both the branchiopods (Briggs, 1981; Briggs *et al.*, 1992) and phyllocarids (Walcott, 1912; Henriksen, 1928; Wills *et al.*, 1994, 1995), but also has many

highly plesiomorphic attributes (Briggs, 1981). Similarities between the bivalved Canadaspidida (*Canadaspis* and *Perspicaris*) and the phyllocarids have been noted by numerous workers (Novozhilov, 1960; Walcott, 1912; Raymond, 1920; Henriksen, 1928; Schram, 1986). However, their position remains controversial (Dahl, 1983, 1984; Bergström, 1992), much depending on the interpretation of limb morphology (Briggs, 1992). *Lepidocaris* from the Middle Devonian Rhynie Chert has been allied with the anostracan branchiopods (Scourfield, 1926; Sanders, 1963; Walossek, 1993) and the brachypodans (Schram, 1986).

TYPES OF DATA

The present study sought to compare two independent sources of data: (1) the distribution of morphological character states across extant and extinct taxa, and (2) data on the first appearances of taxa in the fossil record. These two classes of information are independent. Character data (1) have two applications: (1) they form the basis for estimates of phenetic proximity and morphological disparity; (2) in conjunction with hypotheses about character polarity (from outgroup comparison or ontogeny), character data can be used to infer phylogeny.

The uses of stratigraphic data in phylogeny reconstruction

Attitudes to the use of stratigraphic data in phylogeny reconstruction vary considerably (Janvier, 1984; Hallam, 1988). At one extreme, stratopheneticists derive morphological clusters (i.e. metaspecies; Smith, 1994) at numerous successive time intervals, and then join the most similar clusters in adjacent intervals to directly map out an evolutionary tree along the time axis (or up the stratigraphic column) (e.g. Gingerich, 1979; Fordham, 1986). At the other extreme are cladists (see Patterson, 1981) who (if they incorporate data from fossils at all) consider the stratigraphic distribution of taxa an irrelevance. The middle ground is occupied by workers who variously utilize stratigraphic data in a cladistic context. Four approaches have recently received most attention:

Polarizing characters

This rests on the assumption that the oldest known character states are the most plesiomorphic, and requires a virtually complete fossil record for the groups of interest. A more conservative ('fine tuning') example of this approach is provided by Fortey & Owens (1990).

Stratocladistics

The order in which groups appear in time (and therefore in the fossil record) should correlate with the order of cladistic branching. An irreversible character coding the relative timing of group origins in the record is analysed alongside the morphological data, thereby adding a 'stratigraphic debt' to the total tree length (Fisher, 1994). The method selects the shortest tree or trees overall (which might differ from the most parsimonious tree(s) based solely on an analysis of morphology).

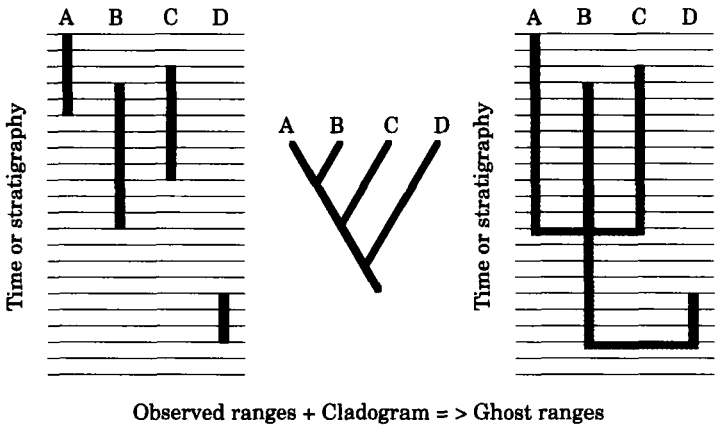


Figure 1. Simple illustration of how observed stratigraphic ranges (black) can be joined by the nodes of a cladogram to infer the existence of ghost ranges (grey).

Wagner's method

Wagner (1995) developed a parsimony-based method for detecting probable ancestors in species-level phylogenies (at the expense of preferring trees necessitating more morphological state changes). Information on the known stage distribution of species (rather than just times of first occurrence) are used to calculate error bars for probable ranges (e.g. Strauss & Sadler, 1989; Marshall, 1990, 1994). Putative sister group relationships are tested, and cladograms containing them rejected where the error bars do not overlap. Where error bars overlap and the older taxon is a species (rather than a clade), the species may be hypothesized to be an ancestor (and 'reversed' steps added to the tree length). Overlapping ranges necessitate no action.

Stratigraphy as an ancillary criterion for choosing among equally parsimonious trees

The morphological data set alone is used to generate the set of most parsimonious trees. A variety of methods [e.g. stratigraphic debt (Fisher, 1994), Stratigraphic Consistency Index (Huelsenbeck, 1994), Relative Completeness Index (Benton, 1994)] can then be used to calculate the 'goodness of fit' of all the trees to stratigraphy. Stratigraphy acts as an ancillary criterion, and generates no new topologies.

Identifying ghost ranges

If a cladogram is derived with no reference to stratigraphy, then nodal branching order can be compared with the sequence of appearance in the fossil record as an independent data set . This approach underlies recent work identifying ghost ranges and testing the quality of the fossil record (Huelsenbeck, 1994; Benton, 1994, 1995; Benton & Hitchin, 1996; Hitchin & Benton, 1996). If the record were complete, sister groups would appear simultaneously in time. Where one taxon appears earlier than its sister, the latter is hypothesized to have a ghost range extending to the origin of the former (Fig. 1). Excellent examples of the application of this approach

are provided by Glenner et al. (1995) (cirripede evolution), and Dunlop & Selden (1998) (the phylogeny of fossil chelicerates).

Such methods make several assumptions, including: (1) taxa are monophyletic; (2) the cladogram is correct and (3) there is no significant anagenesis within taxa. The first is a more general assumption of the cladistic method, and subject to the same strictures. The second is axiomatic, and can only be substantiated under the most controlled conditions (never when dealing with fossils). The third assumption is more problematic. It requires that taxa first appear in the record with all their diagnostic features, which may be an oversimplification (Shultz, 1994). Evidence suggests that most morphological change is associated with cladogenesis as discrete events, punctuated by periods of stasis or equilibrium (Gould & Eldredge, 1977). However, it is possible for groups to diverge with little or no immediate changes in morphology, differences gradually accumulating in one or both lineages over time. Hence, early fossil representatives of a taxon might not be recognizable as such. If a cladogram is assumed to be correct, comparison with the fossil record can set a lower bound for the origin of bodyplan morphologies. This pushes the inferred history of lineages back as far as the fossil record and cladogram will justify. Importantly, this is not an absolute lower bound, since the fossil record of sister taxa need not extend back to their actual cladogenesis. One can never be sure one has found the oldest fossils of a given group.

METHODS

Stratigraphic data

The fossil record of crustaceans is far from uniform. Despite their abundance throughout the Phanerozoic, the records of several groups are erratic. A few heavily-mineralized taxa (e.g. decapods and ostracods) account for the majority of fossil specimens and species. However, most crustacean taxa are 'soft-bodied' and known from rare, temporally and environmentally-scattered Konservat Lagerstätten (Seilacher, 1990). There are fewer marine-derived Lagerstätten after the mid-Palaeozoic, and these are correspondingly less important as sources of information. Such 'taphonomic windows' alone cannot yield a complete picture of evolution at the genus or species level, neither can they provide a continuous record of the temporal pattern of diversity or environmental distribution. Moreover, there are no simple metrics by which to correct for such preservational biases. A study of *bodyplan* disparity through time is a more realistic proposition. The fossil records of bodyplans, like those of higher taxa, are less susceptible to variations in the quality of preservation or the completeness of the fossil record. Range data have been taken primarily from Briggs *et al.* (1993) and Whatley *et al.* (1993).

The morphological data base

Taxa

The intention is to define a morphospace from a representative sample of crustacean bodyplan variety (see analagous studies for all arthropods; Briggs *et al.*,

1992, 1993; Wills *et al.*, 1994, 1998). One exemplar was coded from each crustacean order (and suborders in some cases), according to the classification scheme of Schram (1986). Using his taxonomy ensured adequate coverage of crustacean morphological variety (most of Schram's orders are recognized as subclasses, orders or suborders by most other workers). Where the relationships of taxa within these groupings was uncontested, a primitive or basal representative was coded, and where no such consensus existed, genera with a 'representative' or less specialized morphology were used.

It would be impracticable to code all of the genera observed at each horizon. Some orders would be represented by numerous exemplars, others by just single species (which would potentially bias a morphospace study by emphasizing smaller variations in the more richly sampled group). Moreover, coding many of the oldest fossils (which can be fragmentary but unambiguous exemplars of their orders) would necessitate the incorporation of large amounts of missing data.

Although many Cambrian taxa are are known only from fragmentary specimens (often lacking appendages), most of those coded in the present data base have been described in considerable detail. There are five genera from the Burgess Shale. The Canadaspididae (*Canadaspis* and *Perspicaris*) have been monographed (Briggs, 1977, 1978, 1992), and the position they occupy in morphospace is well constrained (despite some mobility on cladograms) *Odaraia* has been described in similar detail (Briggs, 1981) [older treatments are provided by Størmer (1944) and Rolfe (1959)], while work on *Waptia* (Briggs, 1977) has not been superseded in over 20 years (examination of the specimens confirms that some details are obscured, with some consequent missing data). Many workers believe that *Priscansermarinus* was a pedunculate barnacle (e.g. Collins & Rudkin, 1981; Schram, 1982; Hessler *et al.*, 1982), although Briggs (1983) expresses some misgivings. The remaining taxa have been described from exquisitely-preserved phosphatized material. *Martinssonia* (Müller & Walossek, 1986), *Skara* (Müller, 1985), *Bredocaris* (Müller & Walossek, 1988) and *Rehbachiella* (Walossek, 1993) from the Upper Cambrian 'Orsten' fauna have all been exhastively monographed from SEM studies. *Dala* (Müller, 1983) has also been described in considerable detail, although the anterior appendages are unknown, and its entry includes some missing data. *Hesslandona* and *Vestrogothia* have been scored from detailed descriptions (Müller, 1979, 1982), and are coded separately from the ostracods (the Cambrian phosphatocopine 'ostracods' and the Myodocopida/ Podocopida do not resolve as a clade, and other 'ostracod' taxa known only from carapaces have not been included).

Some fossil genera have invited more than one interpretation. In some cases, coding according to alternative reconstructions can result in different resolutions in cladograms (Schram & Hof, 1998). [Inevitably, it is also possible that preconceptions about relationships have biassed the interpretations of specimens (Gould, 1989)]. Alternative interpretations of *Canadaspis* are offered by Dahl (1984) and Xianguang & Bergström (1997). *Nahecaris* (an archaeostracan) has been coded from the interpretation and reconstruction of Bergström *et al.* (1987). *Tesnusocaris* (an enantiopod) is coded from interpretations by Schram (1983) and Schram & Emerson (1986, 1991), rather than the earlier work of Brooks (1955) (who compared the fossil principally with cephalocarids and branchiopods). Whether or not the duplosegmentation hypothesis of Emerson & Schram (1990) is adopted has no consequence for coding the present character set.

Characters

Determining homologies for a group as large and plastic as the Crustacea is highly problematic (Schram, 1986). Many of the specific issues encountered in constructing and analysing a morphological data base have been considered elsewhere (Wills, 1998a). Most discussions of crustacean phylogeny emphasize tagmosis patterns, numbers of appendages, the numbers of podomeres in the rami of these appendages, and the morphological specialization of limbs (e.g. Hessler & Newman, 1975; Cisne, 1982; Hessler *et al.*, 1982; Wilson, 1992; Ito, 1989). Detailed comparisons of appendage morphology are often pivotal in attempts to determine the affinities of enigmatic fossil taxa (e.g. Scourfield, 1926; Briggs, 1992; Müller, 1983). Particular emphasis has been placed on the condition of the first five (cephalic) limbs. Several important fossils express markedly different head appendage morphologies from any modern crustacean, and in a few cases have fewer cephalic appendages than their modern counterparts. Some authors (e.g. Müller & Walossek, 1986) consider these differences sufficient to exclude some fossil taxa from the crustacean crown group. The present data base codes the morphology of all five anterior appendages independently (cf. Wills *et al.*, 1994, 1995, 1998 for all arthropods). The first four post-maxillary appendages have also been coded, a number intended to capture most of the variation observed in anterior trunk appendage morphology throughout the crustaceans. The resulting data base is composed of 135 characters coded for 62 taxa. A character matrix and a list of characters and codings can be downloaded from http://palaeo.gly.bris.ac.uk/personnel/Wills/data.html.

Cladistic analysis

A thorough cladistic treatment of a similar morphological data base is provided by Wills (1998a), and the reader is referred to that source for a full discussion of ordering and weighting procedures. In summary, however, characters relating to numbers of podomeres, tagmata and other organs have been ordered (considering them as internested sets of synapomorphies), while characters relating to numbers of podomeres (often with very large numbers of states) have also been ranged (downweighted such that transitions between extreme states add a length of 1 to the cladogram). The second to fourth thoracopods (appendages 7–9) are homonomous in most entomostracans, but variously modified for trophic functions in many eumalacostracans. In order to preserve informative variation in the latter group [while avoiding the pitfalls of duplicating data (serial homologies) and 'overweighting' for other groups] characters relating to appendages 7–9 have been scaled to a third of the weight for other characters (using a base weight of 1000). For a summary of the treatment of each character, see the PAUP file at the above web site.

PAUP 3.1.1 (Swofford, 1993) was used to find the most parsimonious trees, using simple, closest, and 500 random additions of taxa followed by TBR branch swapping. The resulting networks were pulled down with the remipedes (Enantiopoda and Nectiopoda). As discussed above, remipedes have consistently emerged as the most basal crustaceans in all analyses of the arthropod clade by the author (e.g. Briggs *et al.*, 1992; Wills *et al.*, 1994, 1995, 1998. See also Emerson & Schram, 1998; Schram, 1982; Zrzavy, Hypsa & Vlàsková, 1998). The object here is to infer phylogeny within the crustaceans (not the position of the crustaceans within the arthropods). Continuing uncertainty over the identity of the crustacean sister group (compare

Emerson & Schram, 1998; Spears & Abele, 1998; Wheeler, 1998; Wills *et al.*, 1998; Zrzavy *et al.*, 1998) means a remipede rooting is considered the most reliable way to polarize characters within the rest of the crustacean clade. All of the shortest trees from the PAUP runs were saved, and a strict consensus derived (Fig. 2).

Assessing the degree of stratigraphic congruence

The strict consensus of the most parsimonious cladograms (24) has been plotted against the fossil record, revealing extensive ghost ranges (Fig. 3). This naturally begs the question of how extensive the ghost ranges are, and how well the cladistic branching order matches the order of appearance of the fossils. Two indices for assessing 'gappiness' and the 'goodness of fit' of cladograms to the fossil record are in common use: the Relative Completeness Index (RCI) of Benton (1994) and the Stratigraphic Consistency Index (SCI) of Huelsenbeck (1994). A third, the Gap Excess Ratio (GER) is briefly discussed here.

The Relative Completeness Index

The RCI is calculated as 1, minus the sum of all ghost ranges (the Minimum Implied Gap or MIG) over the sum of all observed ranges (the Standard Range Length or SRL), multiplied by 100% (Fig. 4B). However, the distribution of observed stratigraphic ranges for any given set of taxa determines an upper and lower bound for the *possible* amount of ghost range. For example, if all taxa originate at the same time, the maximum and minimum ghost range will both be zero. If half of the taxa originate in the Recent and half in the Cambrian, then the minimum ghost range will be the length of the Phanerozoic (one link connecting the two groups). The maximum ghost range in this situation will be equal to the minimum ghost range, times the number of Recent taxa. Tree balance further constrains the distribution of possible RCI values (Siddall, 1996).

The Stratigraphic Consistency Index

The SCI expresses the number of nodes in the cladogram that are consistent (have an older or equally old sister group) as a fraction of the total number of nodes (Fig. 4A). However, this measure is particularly sensitive to the location of the oldest taxa (Hitchin & Benton, 1996). For example, if the branching order from the base to the tip of a pectinate cladogram matches the order of appearance of the taxa in the record, then the SCI will be at its maximum (1.0). If just the basalmost and one of the uppermost taxa are swapped, SCI collapses to its minimum (0.0). This is true even if the implied ghost ranges throughout the cladogram are very small (nodes are either consistent or inconsistent).

The Gap Excess Ratio

While the SCI merely provides an index of the proportion of consistent nodes in a tree, the RCI conflates a measure of the extent of implied ghost ranges with the extent of observed ranges. Benton *et al.* (1998) demonstrated that this is a desirable property of the RCI when attempting to assess the completeness of the fossil record. However, it does mean that the RCI is not a pure index of stratigraphic congruence *per se*. A new index that expresses just the proportion of the total ghost range

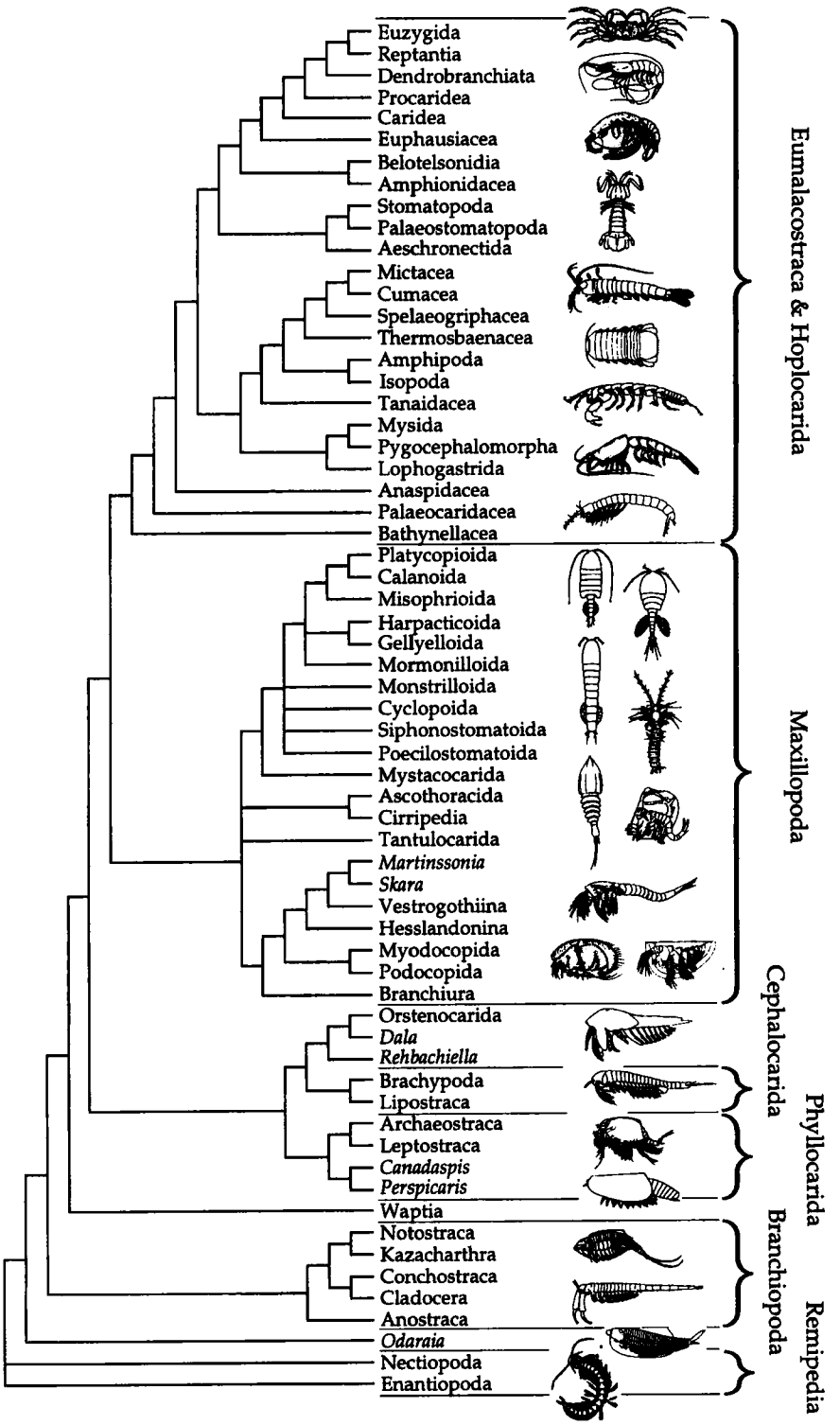


Figure 2. Strict consensus of 24 most parsimonious trees resulting from analysis of Recent and fossil crustacean taxa. Icons represent orders (not necessarily the representative coded).

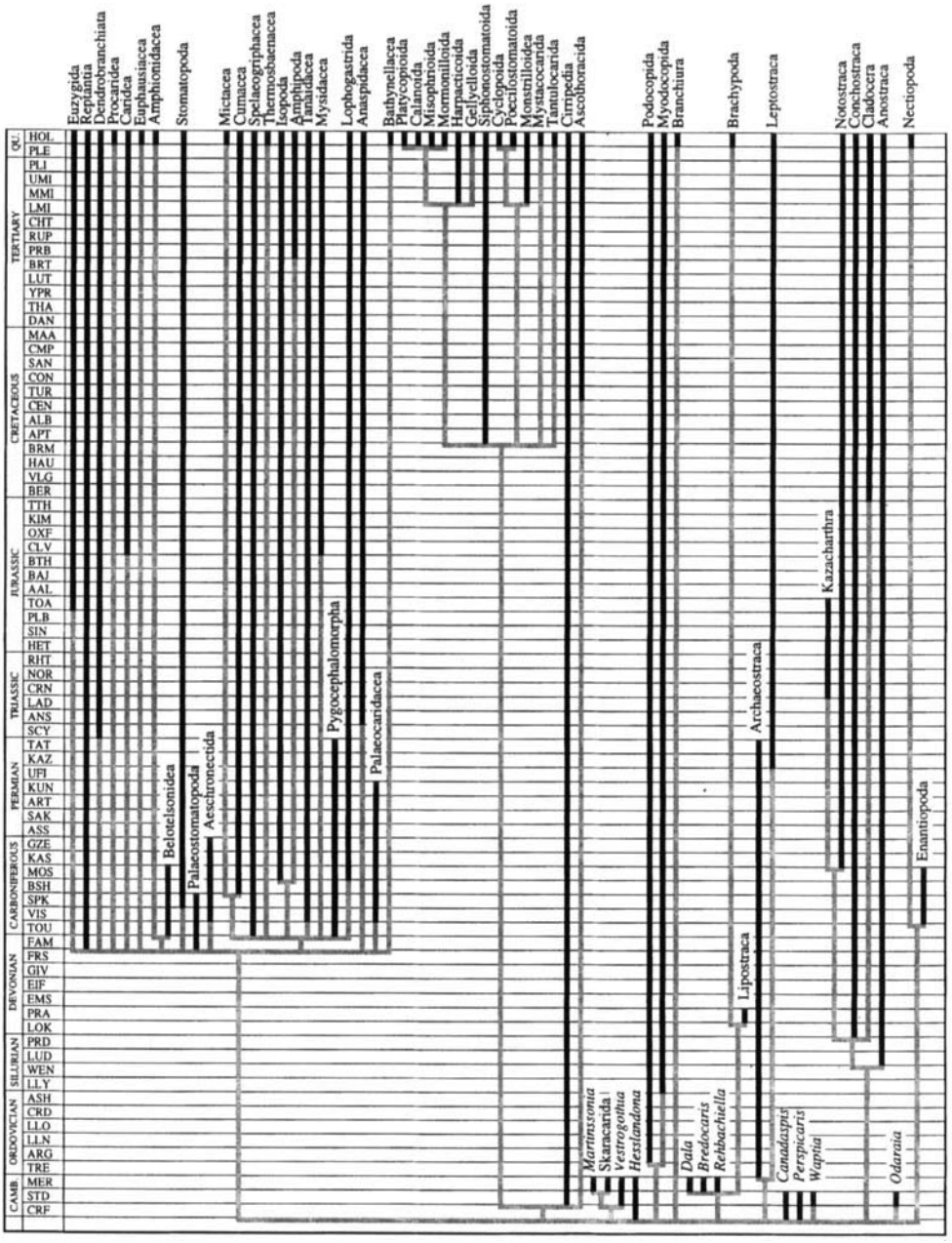


Figure 3. Observed ranges (black) and cladistically-implied ghost ranges (grey) for Recent and fossil crustacean taxa.

necessitated by the constraints of a cladogram is the Gap Excess Ratio (GER) (Fig. 4) (Wills, 1998c). This is given by:

$$GER = 1 - (MIG - G_{min})/(G_{max} - G_{min}) \quad (1)$$

MIG is the 'Minimum Implied Gap' as calculated for the RCI. G_{min} is the 'Minimum Gap' (the minimum possible sum of ghost ranges for any given distribution of origination dates). The Minimum Gap is observed in a completely pectinate tree, with taxa ordered from youngest to oldest, crown to base (Fig. 4C). More simply, G_{min} is given by the difference in origination times between the oldest taxon and the youngest taxon. G_{max} is the 'Maximum Gap', the maximum possible sum of ghost ranges for a given distribution of origination dates. This is given by the sum of the differences in origination times between the oldest taxon and every other taxon (Fig. 4D).

GER values range from 0.00 (the worst possible fit, when MIG is equal to G_{max}) to 1.00 (the best possible fit, when MIG equals G_{min}). GER values of 1.00 are possible even for point occurrences of fossils, which will yield large and negative RCI indices. The GER scores over the SCI and RCI since it enables the comparison of stratigraphic congruence for cladograms referenced to different distributions of range data. However, like the RCI and SCI, the GER provides no control for the theoretical constraints of tree balance.

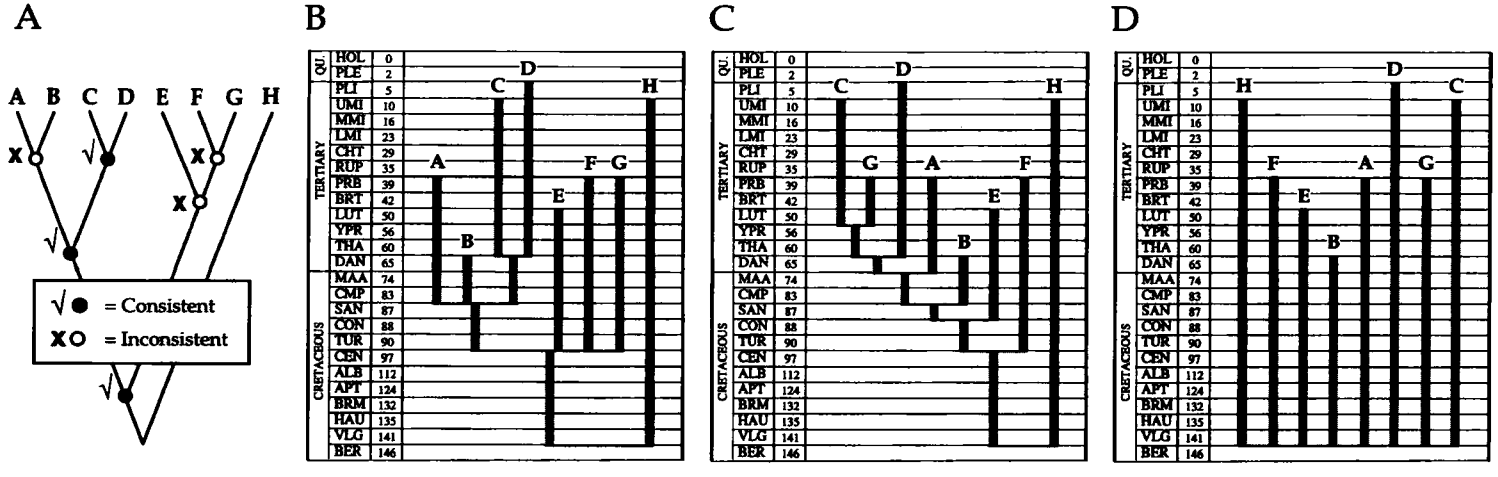
Randomization tests

None of the indices discussed above provide *tests* of stratigraphic congruence. They merely *describe* aspects of congruence, but in a manner that prevents direct comparison with other trees (although the GER is the most suitable for this purpose). Randomization tests determine whether congruence is significantly better than expected by chance, controlling for differences in tree size, shape, and the distribution of range data. The resulting scores are independent of context, and can be compared readily between trees. They operate by randomly permuting the observed stratigraphic range data across taxa a large number of times (e.g. 1000), while holding cladogram topology constant (Huelsenbeck, 1994). RCI, SCI and GER values are recalculated for each randomization, yielding a distribution for each index. The percentage of randomized values worse than or equal to that derived from the original data provides a measure of the uncertainty that a given congruence index is better than random. If this percentage is small (say, 5%), then the fit of the cladogram to stratigraphy is significantly better than would be expected by chance. A program for testing RCI, SCI and GER values in this manner given one or more PAUP, MacClade (Maddison & Maddison, 1992) or user-typed tree files and range data is available from http://palaeo.gly.bris.ac.uk/personnel/Wills/Gho2.html.

Phenetic analysis

Principal coordinates and other ordination methods

The cladistic analysis uses a mixture of character types (ordered and unordered multistate, variously weighted and ranged). The intention was to analyse the same data phenetically in an analogous manner. This is not possible with straightforward



Observed tree

SCI = 3/6 = 0.50

Observed tree and distribution of ranges
Standard Range Length (SRL): 386
Minimum Implied Gap (MIG): 173
RCI = (1-(173/386)) x 100% = 55.2%
SCI = 3/6 = 0.50
GER = 1-(173-112)/(523-112) = 0.85

Tree with smallest possible MIG (G_{min})
Standard Range Length (SRL): 386
Minimum Implied Gap (MIG): 112
RCI = (1-(112/386)) x 100% = 71.0%
SCI = 6/6 = 1.00
GER = 1-(112-112)/(523-112) = 1.00

Tree with largest possible MIG (G_{max})
Standard Range Length (SRL): 386
Minimum Implied Gap (MIG): 523
RCI = (1-(523/386)) x 100% = -35.5%
SCI = 0/6 = 0.00
GER = 1-(523-112)/(523-112) = 0.00

Figure 4. Examples of SCI, RCI and GER calculation (see text). A, observed tree (with SCI calculated according to the distribution of ranges in 'B'). B, observed tree and distribution of stratigraphic range data, yielding an RCI of 54.6%. GER is derived from the G_{min} and G_{max} values calculated in 'C' and 'D'. C, stratigraphic ranges from 'B' rearranged on a pectinate tree to yield the smallest possible MIG or G_{min}. GER is 1.00 by definition. D, stratigraphic ranges from 'B' rearranged on a pectinate tree to yield the largest possible MIG or G_{max}. GER is 0.00 by definition.

factor analysis. Simple principal components analysis (PCA) derives either a covariance or correlation matrix between variables, and then extracts component scores for taxa on each new axis by scaling them in proportion to either the corresponding eigenvectors, or to unity. The net effect is that all variables (characters) must be treated as continuous (ordered) (e.g. Briggs *et al.*, 1992). Wills *et al.* (1994) described a method for unordering and weighting characters that overcomes this problem by recoding the matrix. However, this procedure is a little time-consuming, and the data base must be recoded every time a change is made to the weights or the characters and the taxa included (impractical if one wishes to investigate the effects of different treatments of the data). An alternative approach (particularly if one is not interested in the contribution of each character to each abstracted axis) is the use of principal coordinates analysis (PCO). PCO extracts axes of greatest variance from any matrix of inter-taxon distances (or similarities). A PCO of Euclidean distances yields the same configuration of points as a principal *components* analysis (PCA) (using a covariance matrix, and scaling components to their corresponding eigenvectors). However, if the distance matrix is not Euclidean, a greater or lesser amount of distortion (in the form of negative eigenvectors and associated axes that cannot be interpreted) usually results. Unfortunately, most procedures that derive Euclidean distance matrices treat all characters identically (either ordered or unordered), while methods that allow the use of mixed character types yield non-Euclidean distances [these are suitable for producing dendritic clusterings (phenograms), but not as the basis for morphospace plots]. Gower's (1966) generalized similarity coefficient utilizes mixtures of ordered and unordered characters, but the contributions of each character to each inter-taxon distance are simply averaged, and the matrices are not Euclidean. The solution implemented here was to use an approach similar to that of Gower to derive a more flexible, generalized Euclidean distance coefficient. For this study, three types of characters were recognized: (1) unordered multistate, (2) ordered, and (3) ranged. For a given comparison between two taxa (i and j), unordered characters contribute a univariate distance of 1.00 in all cases where states do not match, and zero where states do match. Ordered characters contribute a univariate distance equal to the difference in their numerical scores. Ranged characters contribute a univariate distance equal to the difference in their numerical scores, expressed as a fraction of the maximum and minimum scores for that character. All of the univariate distances are squared, and the squared distances added. The square root of this sum provides a Generalized Euclidean Distance metric (GED) suitable for PCO ordination. Characters can be weighted by duplicating them in the calculation.

If i and j are two taxa in the data matrix, and k is the data for the kth variable or character, then X_{ik} is the datum for the kth character of taxon i. GED will then be given by:

$$\text{GED}_{ij} = \sqrt{\Sigma_k W_{ijk} S_{ijk}^2} \tag{2}$$

where W_{ijk} is an integer (usually 1), indicating the character weight, and S_{ijk} is the character distance for the comparison. The formula for S_{ijk} depends on the character type.

For unordered multistate characters, $S_{ijk} = 0$ for matches, and 1 for mismatches.

For ordered characters, $S_{ijk} = \left| X_{ik} - X_{jk} \right|$.

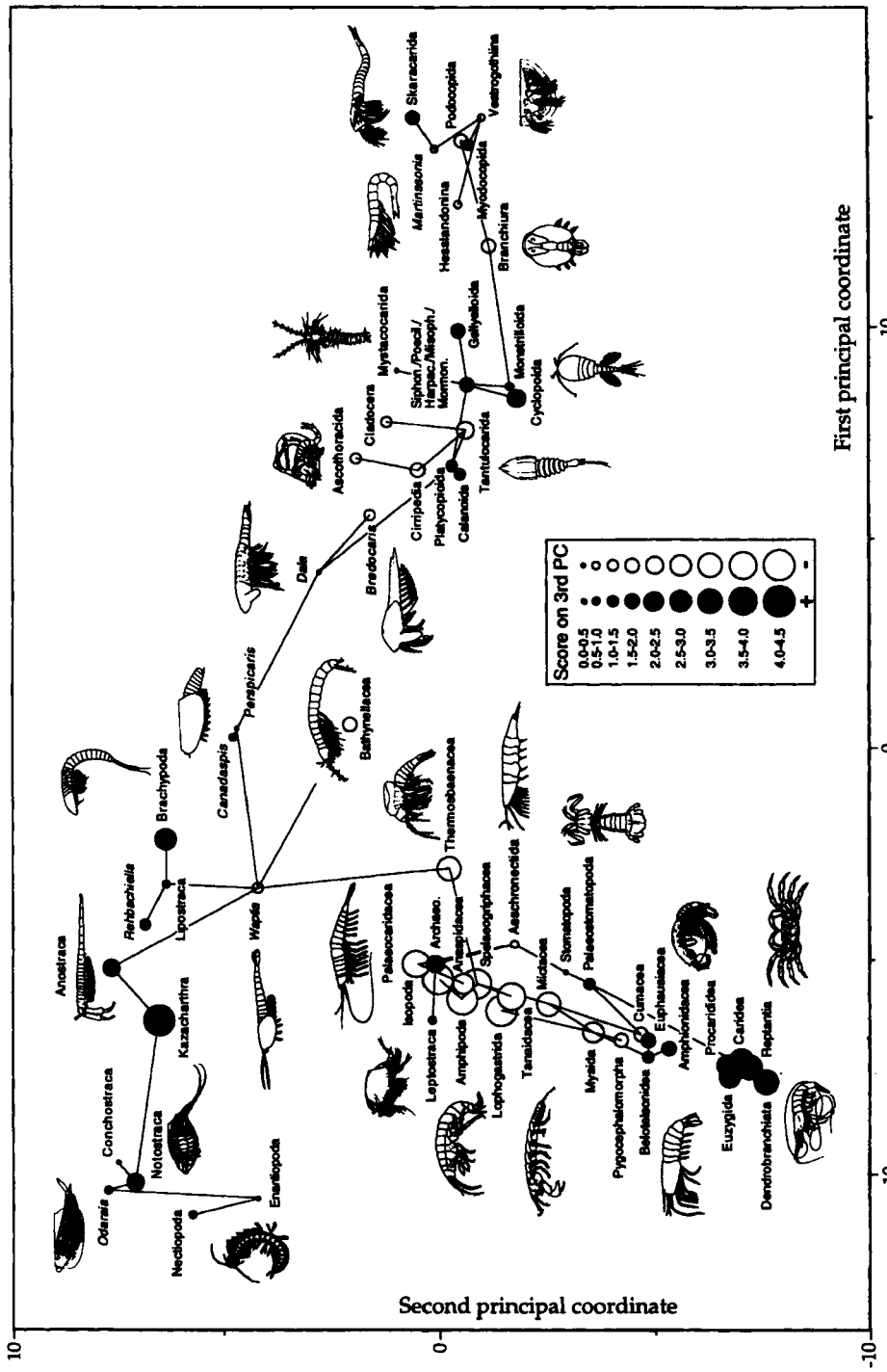


Figure 5. Principal coordinates analysis of fossil and Recent arthropods. Taxa have been plotted with respect to the first three coordinates (third axis in and out of page), which together encompass 75% of the variance in the data set.

For ranged characters, $S_{ijk} = \frac{|X_{ik} - X_{jk}|}{\text{range } (k)}$.

Where a character matrix contains missing entries (common in cladistics), some values of S_{ijk} (the univariate distances) cannot be calculated because data for i or j or both is absent. Where the proportion of missing data is relatively small, these incalculable S_{ijk} values can be assigned weighted 'mean' values, based on the calculable S_{ijk} values. A weighted mean fractional univariate distance for the calculable comparisons ($\overline{S_{ijk}}$) is given by:

$$\overline{S_{ijk}} = \frac{\Sigma S_{ijk} W_{ijk}}{\Sigma S_{(ijk)\max} W_{ijk}} \quad (3)$$

$S_{(ijk)\max}$ values are equal to 1 for unordered and ranged characters, and equal to the range of k for ordered characters. 'Missing' S_{ijk} values in equation (2) are then given by $\overline{S_{ijk}}.S_{(ijk)\max}$.

A Macintosh program ('Matrix') to derive triangular GED matrices suitable for reading into the statistical package MVSP (Kovach, 1990) is available from http://palaeo.gly.bris.ac.uk/personnel/Wills/Mat.html.

A generalized Euclidean distance matrix was calculated from the cladistic data set, and ordinated using the PCO procedure of MVSP (Kovach, 1990). The first 51 of the resulting 63 coordinate axes had positive eigenvalues, leaving less than 2% of the total variance in the data associated with uninterpretable axes. Distortion was therefore trivial. Taxa have been plotted with respect to the first three principal coordinates (Fig. 5), which encompass approximately 75% of the variation in the data. Taxa have also been joined by a minimal-spanning tree (MST) (Gower & Ross, 1969; Rohlf, 1973) based on the same GER matrix. The MST is useful for visualising the location of taxa in all dimensions, given that only the first three can be plotted on paper.

Indices of morphological variation

Four indices of morphological variety were calculated. The first 13 coordinate axes (in this case encompassing 90% of the total variance) were used to calculate products of ranges and variances, and all the axes with positive eigenvalues (here the first 51) were used to calculate sums of ranges and variances. Wills *et al.* (1994) discussed the relative merits of these as indices of disparity. Variances are most appropriate for determining the average dissimilarity between forms, and are relatively insensitive to differences in sample sizes (but are influenced by inconsistencies in taxomomic practice). The total extent of bodyplan variation (disparity) and the filling of extremes of morphospace is best assessed using range-based measures. As discussed in the next section, smaller samples will tend to define a smaller range than larger samples, but these differences are pertinent to a study of bodyplan diversification. Range-based indices are therefore more appropriate as disparity indices, and the sum of the ranges is preferred (see Wills *et al.*, 1994).

The four indices were first calculated for each stratigraphic series (Cambrian to Carboniferous and Tertiary) and stage (Permian to Cretaceous) for lineages actually known from the fossil record. This included lineages known from fossils above and

below the interval in question. For convenience, this is referred to hereafter as the 'observed' disparity.

The four indices were then recalculated, this time for observed lineages plus those inferred from the cladogram. This is designated the 'implied' disparity for simplicity. In instances where a clade with a ghost range (a) opposed a single taxon or clade with an older first occurrence (b), the member of 'a' phenetically closest to any member of 'b' was used to represent clade 'a' throughout its ghost range. This is a conservative approach (though not the only one possible), including ghost taxa closest to regions of morphospace known to be occupied.

There is an inherent and unavoidable asymmetry in the use of ghost ranges, since they can only ever extend the ranges of taxa back in time. Sister groups must derive simultaneously, but can (and very probably do) go extinct independently. Other methods for extending taxon ranges forward in time mostly require a species-level phylogeny. Even if a satisfactory probabilistic method could be devised to extend confidence intervals above and below known ordinal ranges, cladograms could still be invoked to extend the ranges of sister taxa downwards, but not upwards.

Morphological rarefaction

All other things being equal, a larger sample of taxa will (on average) occupy a greater range in morphospace than a smaller one. Foote (1992a) and Wills *et al.* (1994) describe methods for randomly resampling (rarefying) data sets so as to minimize these effects. Two questions are addressed here. (1) To what extent is the apparent difference in the disparity (sum of ranges) of crustacean groups a function of the number of constituent taxa? (2) Is the disparity of crustaceans at each stratigraphic stage more or less than would be expected for a similar number of suborders drawn from the list of all realized bodyplans? All runs were based on 1000 resamplings, with approximate confidence intervals calculated according to Foote (1992a). A Macintosh program for morphological rarefaction is available via http://palaeo.gly.bris.ac.uk/personnel/Wills/Rar.html.

RESULTS AND DISCUSSION

Phylogeny

Crustacean phylogeny and the relationships of problematic taxa are discussed in some detail by Wills (1998a). However, several issues merit comment here.

(1) Gross topology. Parsimony analysis yielded 24 trees (CI=0.269; RI=0.715; RC=0.197). The strict consensus (Fig. 2) conforms precisely to neither Schram's (1986) model, nor to that of Brusca & Brusca (1990), but incorporates elements of both. The Eumalacostraca and Hoplocarida are highly derived rather than emerging basally (*contra* Schram, 1986). However, the phyllocarids dissociate from the Eumalacostraca and Hoplocarida, falling within a basal, paraphyletic series of phyllopod and 'Orsten' groups. A clade including all the maxillopods emerges as sister group to the eumalacostracans and hoplocarids [rather than opposing the Phyllopoda (Schram, 1986)].
(2) The Maxillopoda (including the Ostracoda and Tantulocarida, but including

Skara and *Martinssonia*) resolve as a clade. They are supported by the presence of 11 post-maxillary body segments, and six somites in the thorax (Dahl, 1956), each bearing an appendage. The abdomen is apodous. Trunk endopods tend to be composed of three podomeres.

(3) Phyllocarids (including the Canadaspidida) oppose a clade of Brachypoda [Cephalocarida and *Lepidocaris* (Schram, 1986)] plus the Orsten *Rehbachiella*, *Bredocaris* and *Dala*. This entire clade opposes the Eumalacostraca/Hoplocarida and Maxillopoda. The other major 'phyllopod' group (the Branchiopoda) forms a clade lower in the tree.

(4) Primitive groups. Anostracans resolve low in the tree, while the cephalocarids resolve somewhat higher.

(5) In addition to the problematica discussed above, the location of *Waptia* between the phyllocarid/cephalocarid and branchiopod clades is notable. *Odaraia* resolves basally above the remipedes, and is clearly highly plesiomorphic.

(6) Within the Eumalacostraca/Hoplocarida, the Syncarida are paraphyletic and basal. The Peracarida resolve as a clade (cf. Watling, 1983), within which the shrimp-like mysids oppose all other taxa. The Eucarida and the Belotelsonidea together comprise a clade in opposition to the Hoplocarida. The hoplocarids are usually considered to be the sister group of the Eumalacostraca (Schram, 1969; Bowman & Abele, 1982; Kunze, 1983; Dahl, 1983; Hessler, 1983; Spears & Abele, 1998), and are placed in a separate subclass by Schram (1986). Forcing this alternative resolution for the hoplocarids (Schram, 1969) (using MacClade), adds 1.7% to the total tree length, but 3.2% to the eumalacostracan/hoplocarid portion alone. [The weighting procedure makes it difficult to equate these percentages directly with unit numbers of state changes. However, the weighted length of the entire most parsimonious trees is 602533, and that with the hoplocarids moved is 612949. The minimum possible weighted length is 166002, and the minimum unweighted length 272 (135 characters, some of them with large numbers of alternative states). The addition to tree length resulting from moving the hoplocarids is therefore equivalent to: $(602533/166002) \times 272 \times 1.7\% = 17$ unit state changes. However, it should also be noted that the cost of the move if all characters are weighted to unity from the outset is zero.] The more derived location for the hoplocarids resolved here finds some support from other morphological and total evidence approaches (including data from 18S and 28S rDNA) (Wheeler, 1998).

Completeness of the fossil record and stratigraphic congruence

RCI values for the 24 fundamental trees are all low (from -8.28 to -15.53; negative values indicate that there is more ghost range than observed range). However, GER values are between 0.61 and 0.58 indicating that while the fossil record may be relatively poor, the fit of the most parsimonious trees to the range data (measured in terms of ghost ranges) is reasonably good. Moreover, none of 1000 permuted range assignments produced RCI or GER values equal to or better than this, indicating that the extent of ghost ranges is very significantly less than expected at random. SCI values for the fundamentals ranged from 0.46 to 0.43, indicating that the majority of their nodes are stratigraphically inconsistent. However, considering just the Eumalacostraca/Hoplocarida, it will be seen that some of the

oldest taxa (e.g. Reptantia, Spelaeogriphacea) are amongst the most derived in their constituent subclades (leading to low SCI values regardless of the consistency of the rest of the tree). It is unsurprising, therefore, that the SCI from the original data does not differ significantly from the random permutations (Hitchin & Benton, 1996).

The phylogeny of Schram (1986) (containing an almost identical set of taxa) was also tested for congruence, with similar results. All indices were comparable to those observed above (RCI = −16.02, GER = 0.57, SCI = 0.40), with none of the 1000 randomized trees having a better RCI or GER (a highly significant result). Again, the deviation of the SCI value from the random distribution was insignificant.

Congruence in the Eumalacostraca and Hoplocarida

Congruence was recalculated for just the eumalacostracan/hoplocarid portion of the tree (24 taxa). Despite a more complete fossil record than the tree as a whole (RCI = 8.08), the GER was markedly poorer (0.06) and the SCI slightly poorer (0.41). RCI, GER and SCI significances based on 1000 permuted range reassignments were extremely poor (99% of these randomized assignments yielded *better* RCI and GER values than the original data). It is useful to consider why congruence within this portion of the tree should be so low.

The most obvious explanation would be a wildly inaccurate cladogram. In this context, it is interesting to look at the goodness of fit for some alternative eumalacostracan/hoplocarid trees [phylogenetic ideas have been remarkably static throughout most of the twentieth century (Schram, 1981)]. The only other cladogram of the whole group based on a parsimony analysis of morphological characters (Schram, 1986) had a lower RCI (1.73) and SCI (0.27), but a slightly higher GER (0.13). None of these values deviated significantly from random. The 'traditional' model (illustrated by Schram, 1981) had a much better RCI (28.60) and SCI (0.42) but a poor GER (0.12), with none of the indices being significant. Trees for the peracarids alone also failed to yield congruence values deviating significantly from random [Fryer (1965), RCI = 55.49, SCI = 0.40, GER = 0.03; Pires (1987), RCI = 34.74, SCI = 0.50, GER = 0.02; Watling (1981), RCI = 53.13, SCI = 0.40, GER = 0.09]. While it is possible that all the published trees are hopelessly inaccurate, this seems unlikely.

A second possible explanation would be a severely patchy fossil record. Apart from the taxa restricted exclusively to the Recent, most eumalacostracan and hoplocarid orders arose in a relatively short time interval from the Late Devonian to the Early Carboniferous. The majority of eumalacostracan bodyplans (as evidenced by their contribution to total disparity) appear to have been deployed in a rapid radiation (Schram, 1981). Fossil collecting over the last hundred years has done little to change this picture. It is thought unlikely that the group would remain static for 350 million years or more, before radiating again in several parallel lineages near the Recent. Hence, it seems probable that several apparently solely Recent groups have (in reality) a long history. This is almost certainly true of the Thermosbaenacea (Schram, 1986), where trans-Atlantic links of closely-related modern taxa (Stock, 1976, 1982; Stock & Langley, 1981) strongly suggest the breakup of an ancient group with a Tethyan biogeography (Maguire, 1965).

This still begs the question of *why* no fossils have been found. Probable sister taxa in similar environments can have hundreds of millions of years between their earliest

fossil occurrences (e.g. Isopoda and Amphipoda, Euzygida and Reptantia, Mysidacea and Pygocephalomorpha). Therefore, limited exposure of suitable sediments is unlikely to be the explanation.

Fortey, Briggs & Wills (1995, 1997) consider cladogenesis in lineages of a small size as a possible explanation for the lack of Precambrian fossils and the apparently explosive radiation of the arthropods and Metazoa *en masse* in the Cambrian. This hypothesis requires an external trigger for size increase in numerous parallel lineages. While the Cambrian may be a special case in the fossil record of the history of life (Gould, 1991), it is difficult to envisage a similar explanation for the appearance of several Recent eumalacostracan groups.

A plausible reason why some groups left no fossils is that numbers of individuals (and species) have always been low. The Procarididea, for example, are represented by just a single living species (*Procaris ascensionis*), discovered relatively recently in the Ascension Islands (Chase & Manning, 1972), and known only from two other locailities in Hawaii. Similarly, the Mictacea are known from just two extant species in two families; one deep sea form (*Hirsutia bathyalis*) described from just two specimens (Sanders, Hessler & Garner, 1985), the other (*Mictocaris halope*) more plentiful, and endemic to marine caves in Bermuda (Bowman & Illiffe, 1985). The order Amphionidacea contains just one living species (*Amphionides reynaudii*; Williamson, 1973), ubiquitous but uncommonly reported from depths of around 2000 m or more. [The absence of euphausiacians in the fossil record is more perplexing. Modern forms range in length from about 4 cm to 15 cm, and krill can reach densities in excess of 1000 animals/m^3 of sea water. Several authors (e.g. Dzik & Jazdzewski, 1978; Schram, 1981) have forwarded candidate fossils, but none of these is particularly convincing (Schram, 1986)]. From this perspective, congruence for the Eumalacostraca has deteriorated with the discovery of new living taxa in the last hundred years, and the situation may get worse before it improves. However, since many of the Recent orders are 'morphological intermediates', and have sister groups with an ancient fossil record, it seems unlikely that the pattern of disparity through time will be radically changed by new fossil discoveries.

Another possibility is that some taxa are ancestral to others, with relatively recent groups being derived from the midst of those with a longer heritage. Clearly, paraphyly in some taxa would make the cladistic treatment of the data (and the ghost range approach) problematic. While this possibility cannot be excluded, all the eumalacostracan and hoplocarid orders are considered clades, and there are few evolutionary systematic scenarios that propose the derivation of one order from another (Schram, 1982, 1986; but see discussion by Dahl, 1992). Again, however, the paraphyly of several lineages (though devastating to apparent congruence values) need only have a slight to moderate effect on disparity profiles. The profiles calculated solely from the observed fossil lineages are consistent with the worst-case scenario in which the majority of orders are paraphyletic.

In fact, none of these explanations is mutually exclusive, and the reality may combine elements of all of them.

Crustacean morphospace

Although not clusters in the strict sense, three broad groupings can be recognized in the PCO plot (Fig. 5) (compare with the cladogram; Fig. 2). (1) The right-hand

region contains predominantly maxillopods, and several other taxa (Orsten forms) with an abbreviated series of trunk appendages. At the extreme right of the maxillopod grouping are the ostracods, *Martinssonia* and *Skara*. More centrally, the copepods cluster tightly, along with the mystacocarid, thecostracans and the tantulocarid. To the left are the Orsten *Bredocaris* and *Dala*. (2) The upper left region of the morphospace contains predominantly 'phyllopodous' taxa (those with 'leaf-like' appendages). The remipedes join the branchiopods on the far left, with the canadaspidids on the far right. Most of the taxa thought to exemplify the ancestral condition by various authors (Remipedia, Cephalocarida, Anostraca) plot in this region of the graph. (3) The mid to lower left region is occupied by Malacostracans. These extend perpendicular to the main axis (the first coordinate) defined by the phyllopods and maxillopods.

Morphology within the Eumalacostraca/Hoplocarida ranges from the entomostracan-like Bathynellacea and Thermosbaenacea, to the caridoid Eukyphida, Dendrobranchiata and Reptantia. The Peracarida (all joined by the MST) includes some taxa (e.g. Thermosbaenacea) morphologically proximate to the primitive syncarids, and others (e.g. Mysidacea) showing striking convergence with the caridoid eucarids. This is particularly noteworthy; even though the character matrix does not code for overall form and 'shape', taxa with grossly similar bodyplans plot proximally in the morphospace. This implies congruence between disparity measured in terms of discrete character data and that derived from analyses of overall shape.

Phenetic proximity does not necessarily imply a close phylogenetic relationship (compare Fig. 5 with Fig. 2). The Leptostraca and Archaeostraca (phyllocarids) are phenetically closest to the Eumalacostraca, although they resolve some distance from them on the cladogram. Phenetic convergence *may* explain the desire to classify the Phyllocarida with the Hoplocarida and Eumalacostraca (Malacostraca) (Claus, 1888; Metschnikov, 1868). By the same token, close phylogenetic proximity need not imply that taxa are immediately adjacent in morphospace. Although the Leptostraca/Archaeostraca and the Canadaspidida form a clade, the canadaspidids are phenetically removed from the other phyllocarids (Fig. 5). By contrast, phenetic distinctiveness (in the form of autapomorphies) may explain why the Hoplocarida are usually regarded as a separate taxon from the Eumalacostraca.

The disparity of major groups

The disparity of several major crustacean groups (not all of them clades) measured as the sum of ranges on the first 13 principal components is illustrated (Fig. 6A). The phyllopods are the most disparate group, but are not a clade. The Malacostraca emerge as the most disparate clade, closely followed by the Maxillopoda and Branchiopoda. [The use of the product of ranges on the first 13 axes gives a similar result, while the incorporation of all components captures more variation within the Phyllocarida (corresponding to a level of disparity between the Eumalacostraca and Maxillopoda – not illustrated)]. The groups illustrated are not of an equal size (Malacostraca are represented by 24 taxa and Maxillopoda by 21, while there are just four phyllocarids), and the effects of these differences were investigated with rarefaction. Figure 6B shows the results of 1000 bootstrappings (resamplings with replacement) of four taxa (the number of phyllocarids) from the larger groups. Upper and lower 95% intervals are calculated according to Foote (1992a). The overall

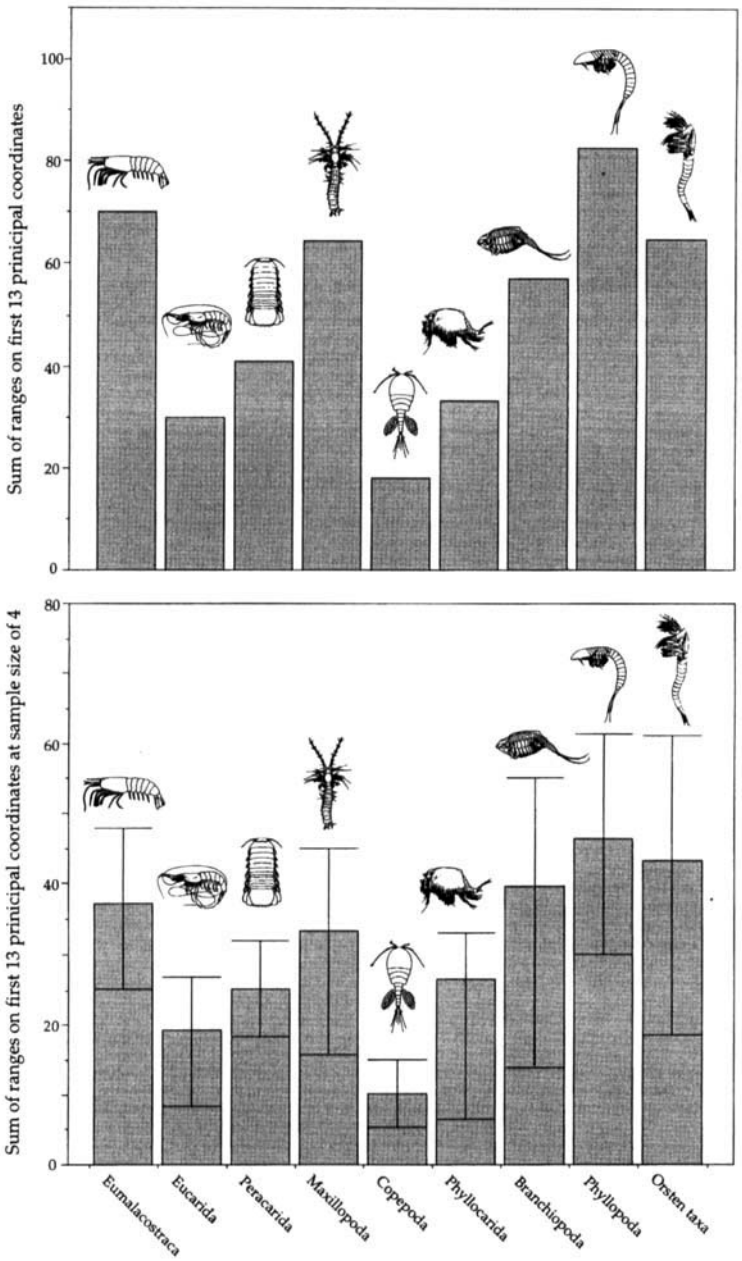


Figure 6. Disparity (sum of ranges on first 13 principal coordinates) for several large crustacean clades and other groups. A, disparity of all taxa in group. B, mean disparity of four taxa bootstrapped from each group (1000 samples, with upper and lower 95% confidence intervals).

trends are similar to those emerging in Figure 6A, but the branchiopods are marginally the most disparate clade.

The five branchiopods display a wide range of bodyplan designs (e.g. compare the abbreviate trunk and non-phyllopodous appendages of carnivorous cladocerans with the long trunk and numerous phyllopods of the notostracans). While the

Notostraca, Kazacharthra, Conchostraca and Anostraca all plot in the top left, phyllopodous region of Figure 5, the Cladocera resolve closer to the maxillopods, and are joined to the tantulocarid by the MST. Within the Malacostraca, the peracarids represent a relatively disparate assemblage [ranging from entomostracan-like thermosbaenaceans, through rather myriapod-like isopods (woodlice) to shrimp-like mysids]. The Eucarida (crabs, lobsters, crayfish, some 'shrimps') are more tightly constrained (Schram, 1986). Within the maxillopods, the copepods (although represented by ten suborders) constitute a very tight cluster, and accordingly have a relatively low disparity (regardless of the measure used, the number of components incorporated, or control for sample size differences). The phyllocarids, by contrast, are represented by just four taxa (not all of them of subordinal status) yet subtend a mean bootsrapped amount of morphospace exceeding the eucarids, peracarids and copepods.

Overall patterns of disparity and morphological variance through time

Sums and products of ranges

Sums and products of ranges provide the most appropriate indices of disparity, and the sum of ranges is preferred (Wills *et al.*, 1994). However, both indices yield similar profiles through time (Fig. 7A,B). The sums and products of ranges for observed lineages in the Lower Cambrian are 34% and 28% respectively of the corresponding Recent values. Sums and products of ranges for observed plus inferred lineages are markedly higher (76% and 80% respectively). Hence, even conservative estimates show the first crustacean fossils to have been widely separated in morphospace, and to have occupied perhaps one third of the Recent hypervolume. Observed disparity increases dramatically through the Cambrian (the range sum in the Merioneth reaches 62% of the Recent, and the range product 59%). Implied disparity also peaks, but much less markedly than observed disparity. The earliest Ordovician sees a drop to the lowest observed levels at any time, while the drop in implied disparity is much more modest. The rest of the period sees little change. Observed and implied profiles show the same pattern through the Silurian and Devonian, with increases in the mid Silurian, early and late Devonian (although increases in observed levels are more pronounced than those in implied levels). Impied disparity reaches its maximum by the beginning of the Carboniferous, and shows little change threafter. Observed disparity shows a gradual increase between the Carboniferous and Recent, though with some perturbations (more marked for the product of ranges). Notable increases occur in the early Triassic, and at the beginning of the Cretaceous.

Sums and products of variances

Variances are most appropriate for measuring the average dissimilarity amongst taxa (Foote, 1992b), rather than the absolute extent of bodyplan variety (disparity). If the distribution of taxa in morphospace were always random, the sum of variances should be largely independent of sample size differences. Similarly, if taxa appeared and disappeared randomly in morphospace through time, sums of variances should remain approximately constant. This is not so for crustacean morphospace.

The observed sum of variances (Fig. 7C) shows irregular growth to a Devonian

peak, then a crash with more moderate oscillation about the mean from the Carboniferous to the Recent. The implied sum of variances (Fig. 7C) shows far more shallow growth into the Devonian, a more modest drop into the Carboniferous, and shallow growth thereafter. The corresponding patterns for products of variances are rather different (Fig. 7D). The observed variance product shows dramatic and irregular growth from the Cambrian to the Devonian, and slight variation about a constant high level thereafter. The curve is similar to those for the observed sums and products of variances (disparity). The implied variance product starts much higher than the observed variance product, reaches a modest peak in the Devonian, and broadly tracks the observed variance product from the Carboniferous to the Recent.

Identical sums of variances can result from very different distributions of variances on successive coordinates (e.g. $3+2+1=6$ and $2+2+2=6$). However, where a large proportion of the total variance sum is invested in a small number of coordinates, the product will always be less than if the total variance were more evenly-distributed over successive coordinates ($3\times2\times1=6$; $2\times2\times2=8$). All coordinates contribute equally to variance products (regardless of the variance they encompass), while they contribute to variance sums roughly in proportion to the corresponding eigenvalues. In the first half of the Phanerozoic, variance sums for observed lineages are higher than variance products because more of the variation is expressed on earlier coordinates. Similarly, the marked Devonian peak in observed variance sums (but not variance products) indicates a probable increase in variance predominantly on the earlier, more important coordinates. Sums and products of ranges tend not to behave in this manner, since successive coordinates are derived so as to encompass successively smaller portions of the total *variance*, with no reference to the range (which can remain stable or increase on later coordinates).

Small sample sizes make it difficult to conclude much from the marked fluctuations of variance from the Cambrian to the Ordovician. The radiation of the branchiopods through the Silurian and the base of the Devonian is responsible for a marked increase in variance-based indices, and the greatest average dissimilarity amongst taxa occurs during the Devonian.

Disparity and the occupation of morphospace through time

The following discussion is limited to range-based indices of morphological variation (the preferred indices of disparity rather than the average dissimilarity between forms). As noted above, sums and products of ranges both yield similar disparity profiles (Fig. 7A,B).

Cambrian

The oldest, well-described crustacean fossils that appear in the Lower Cambrian are phosphatocopines and canadaspidids (Fig. 8A). These have markedly dissimilar morphologies, and while they do not delimit the extremes of crustacean bodyplan variation on all axes, they define a sum of ranges one third of the Recent level. The appearance of several Burgess fossils (*Waptia* and *Odaraia*) proximate to the branchiopod region of the plot serves to increase observed disparity by over 70%

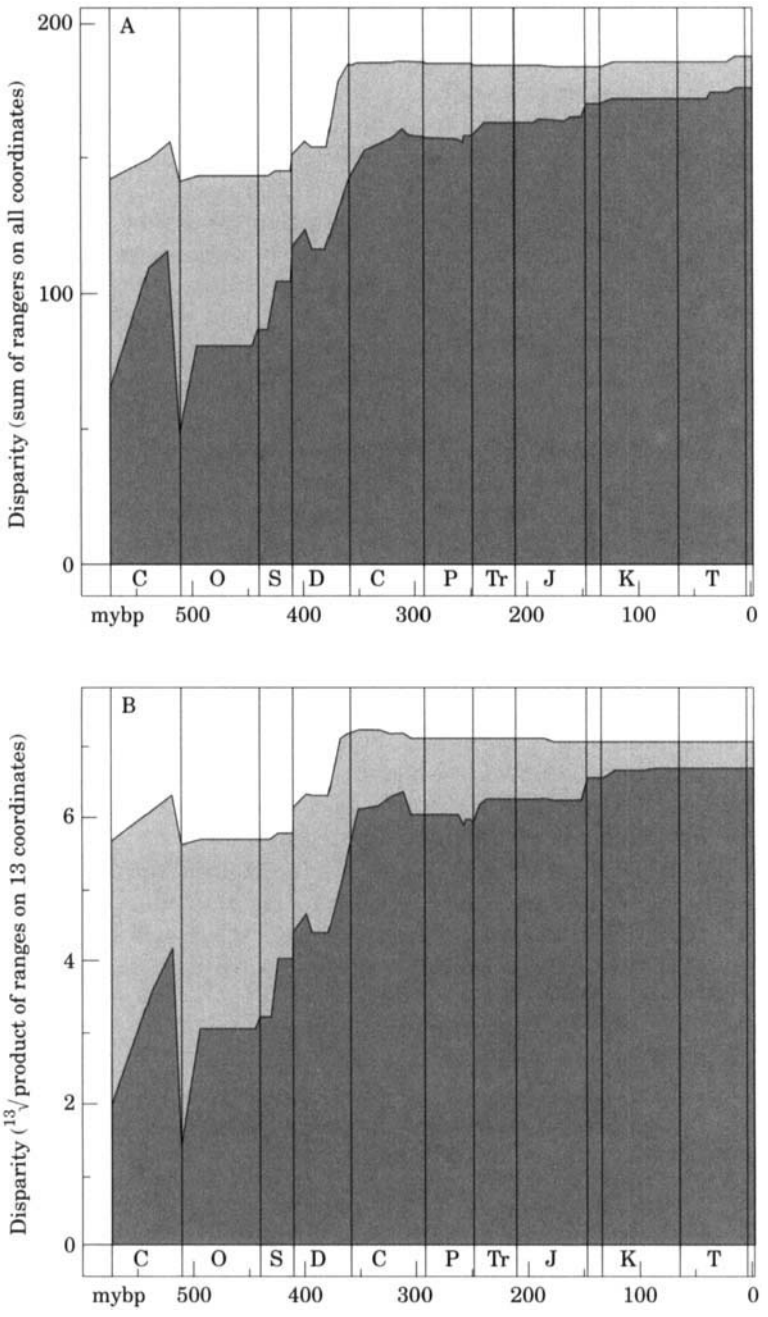


Figure 7. Measures of morphological variation for crustacean groups through the Phanerozoic. A, disparity, measured as the sum of ranges on all principal coordinates (the preferred index). Disparity of observed lineages in dark grey, with the additional disparity implied by ghost lineages in light grey. B, disparity, measured as the 13th root of the product of ranges on the first 13 principal coordinates (encompassing 90% of total variance).

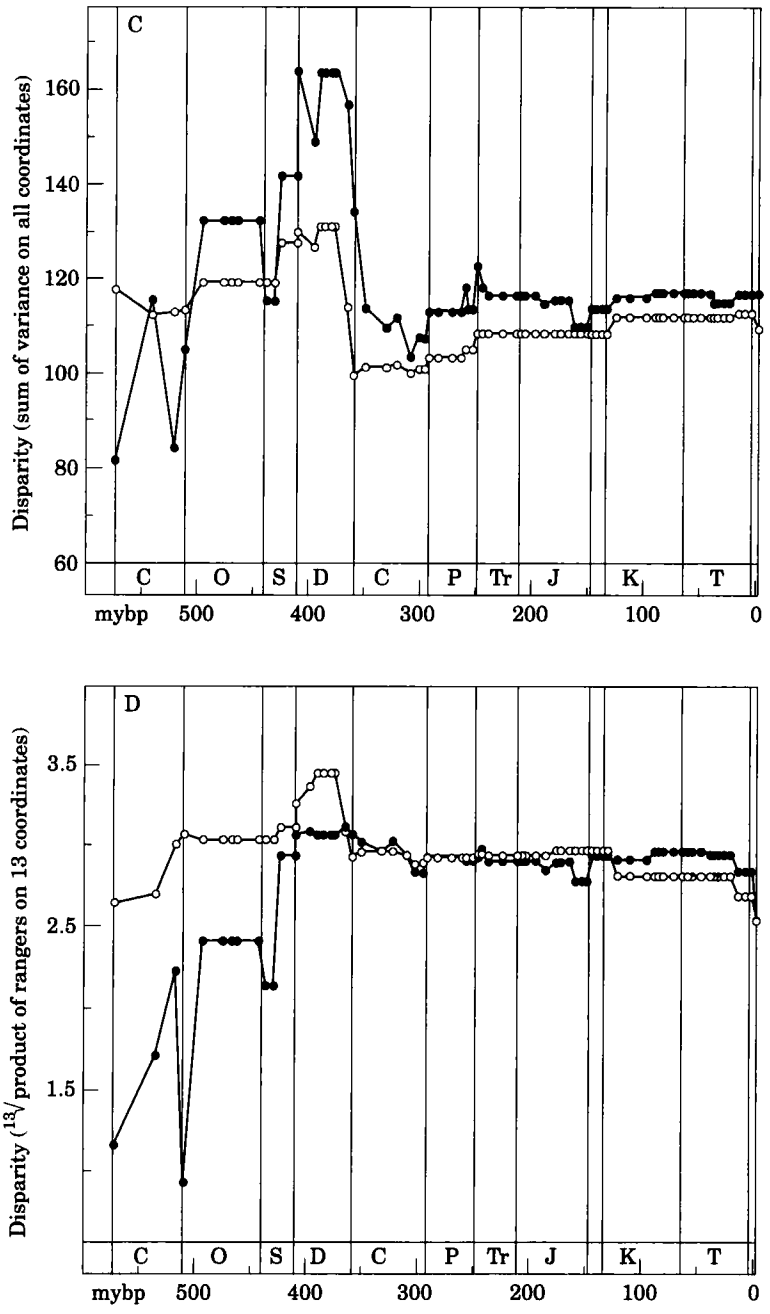


Figure 7—*contd.* C, average dissimilarity between forms measured as the sum of variances on all principal coordinates. D, average dissimilarity between forms measured as the 13th root of the product of variances on the first 13 principal coordinates. Disparity of (●) observed lineages and (○) observed lineages plus ghost lineages.

in the middle Cambrian . The Upper Cambrian Orsten fauna contains taxa close to the maxillopod (*Bredocaris, Dala*) and ostracod (*Skara, Martinssonia*) morphologies, continuing to increase observed disparity to the end of the Cambrian.

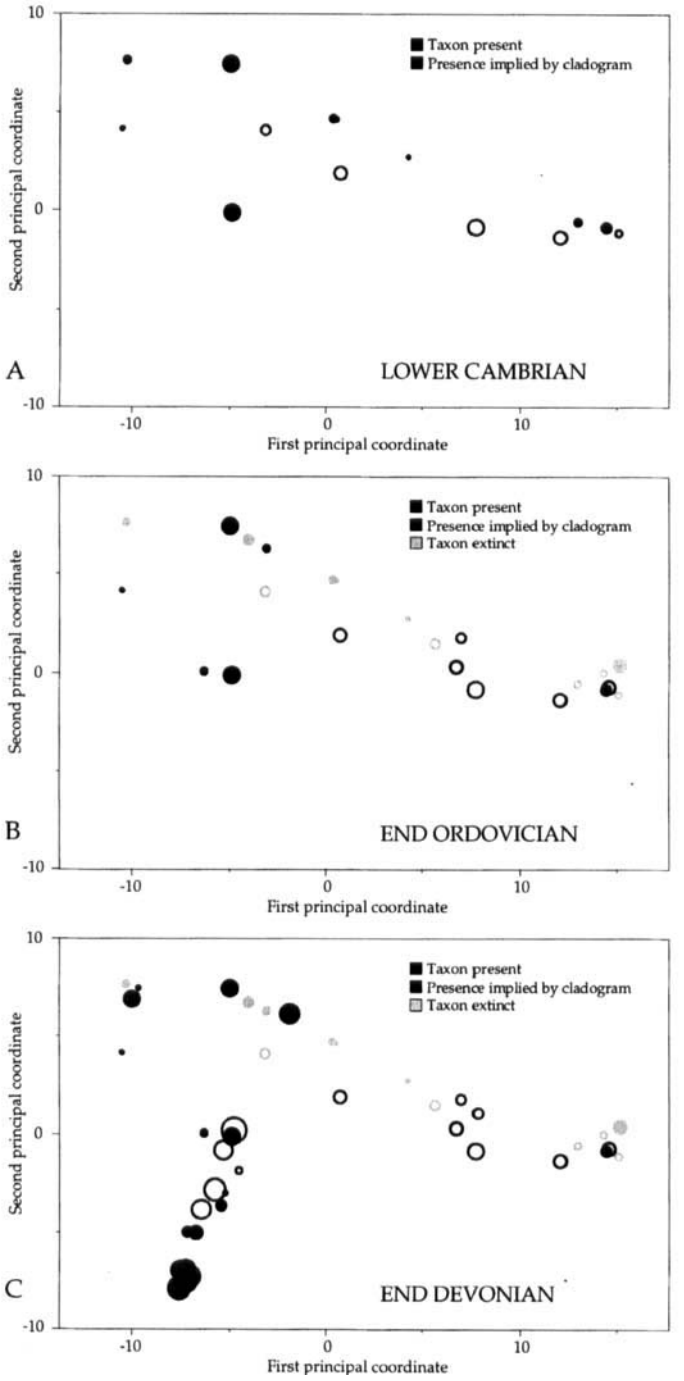


Figure 8. Representative 'time slice' plots of crustacean morphospace. Compare with Fig. 5. First three principal coordinates are illustrated. Third coordinate is at right angles to the plane of the page, filled symbols are positive, open symbols are negative (arbitrary). Black symbols represent taxa with a range extending through the stage illustrated. Dark grey symbols represent taxa with a ghost range. Light grey symbols represent taxa extinct at the stage illustrated. A, Lower Cambrian (Caerfai). B, End Ordovician (Ashgill). C, End Devonian (Famennian). D, End Carboniferous (Gzelian). E, End Jurassic (Tithonian). F, End Tertiary (Pliocene).

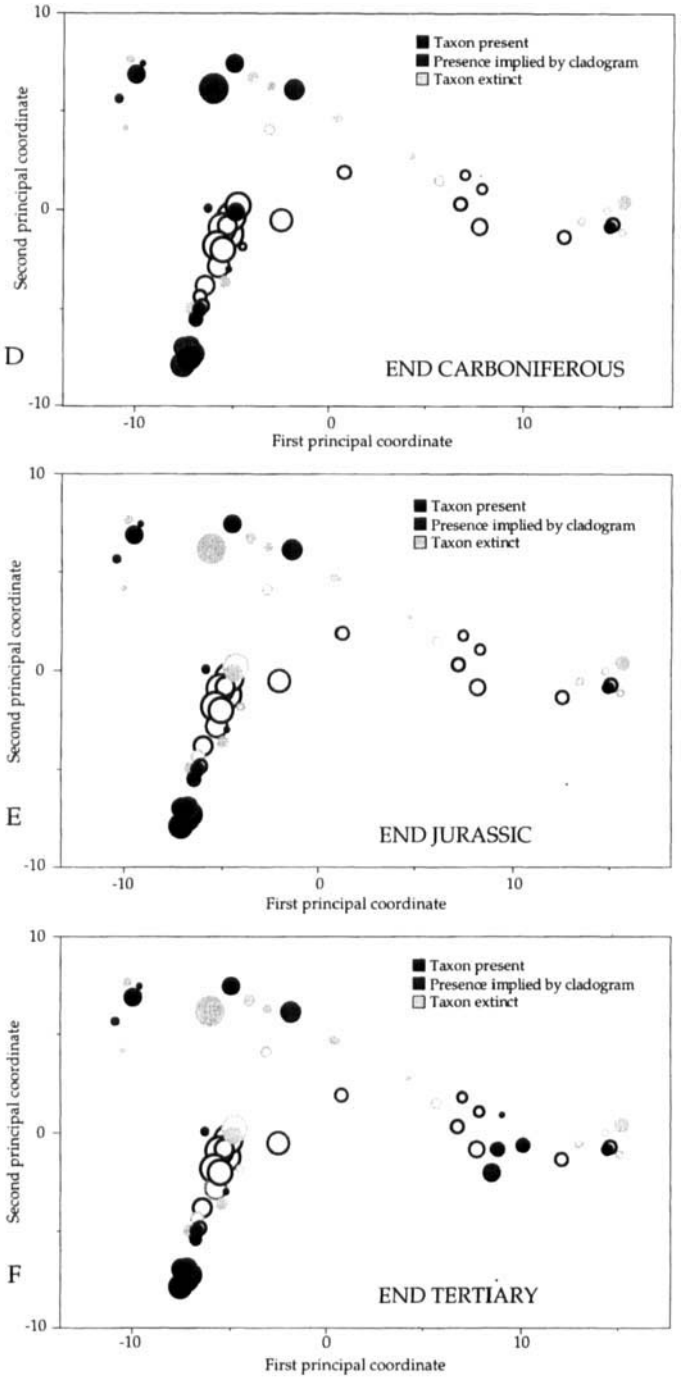


Figure 8—*contd.*

The cladogram implies that most major clades (e.g. Remipedia, Branchiopoda, Phyllocarida, Ostracoda, Eumalacostraca, Copepoda to Thecostraca, *Dala* to Cephalocarida), as well as the genera *Waptia* and *Odaraia*, must have been represented even in the Lower Cambrian. Implied disparity approaches 76% of the Recent level. The implied peak toward the end of the Cambrian is far less pronounced than the observed peak, since some of the previously inferred lineages now appear in the record. Nonetheless, implied disparity reaches 83% of its present level.

Since only well-described fossils are coded from the Cambrian, the indices of Cambrian disparity may be conservative. Numerous Cambrian fossils known only from carapaces have been assigned to the phyllocarids and ostracods, but their status is highly contentious (Schram, 1986) (Cambrian 'phyllocarids' may actually have greater affinities with thylacocephalans). For this reason, the Cambrian portion of Figure 7 should be viewed with some caution.

Ordovician/Silurian

The end of the Cambrian saw the demise of the canadaspidids and all of the Orsten genera. The earliest Ordovician is therefore marked by a decrease in observed disparity to a level below that in the earliest Cambrian, supported only by archaeostracans and cirripedes (Fig. 8B). An increase in observed disparity in the Arenig was maintained throughout the Ordovician. The Silurian saw the appearance of myodocopine ostracods (close to the podocopines, and therefore with little effect) and the anostracan branchiopods [colonizing a more peripheral region of morphospace (upper left in Fig. 5)], bringing observed disparity to over half its Recent level. Numbers of *observed* taxa are small throughout the Cambrian, Ordovician and Silurian, and the extinction of a few lineages has more influence on absolute levels of disparity than at later times. Several regions of morphospace are occupied by single taxa, and their extinction causes the collapse of a large portion of the envelope. The denser occupation later in time means that several taxa or lineages need to become extinct to achieve a comparable effect.

Implied disparity (unlike observed disparity) does not exhibit a catastrophic drop from the Cambrian to the beginning of the Ordovician, and maintains a level similar to that in the Lower Cambrian throughout the Ordovician and Silurian. Again, the inferred presence of several major lineages (Remipedia, Eumalacostraca, Cephalocarida, Copepoda to Tantulocarida) serves to keep the morphospace envelope open.

Devonian

The appearance of conchostracan branchiopods at the beginning of the Devonian restores observed disparity to Middle Cambrian levels. The brief appearance in the Lokhovian of the phyllopodous *Lepidocaris* (Rhynie Chert) (Scourfield, 1926) in a similar location sees another slight disparity increase. The next marked change (Famennian) corresponds to the appearance of the first eumalacostracans, the reptantians and palaeostomatopods (Fig. 8C). Reptantians are a highly-derived group of eumalacostracans (Fig. 2), and plot far from the entomostracan region of morphospace (Fig. 5) (on a remote terminal of the MST). Their appearance represents a significant morphological innovation (implying the existence of numerous ghost lineages for taxa that appear later). The palaeostomatopods are hoplocarids, often regarded as a sister group to the Eumalacostraca (Schram, 1986), but resolved here

as a sister group to the Eucarida plus the Belotelsonidea (Fig. 2). Disparity accordingly rises to its highest level so far.

The pattern of implied disparity is similar to the observed pattern, only displaced vertically. The sum of ranges reaches 95% of its Recent level by the end of the Devonian (Famennian), bolstered by ghost lineages for most eucarid and all syncarid orders, and a diverse branchiopod clade. Peracarids are represented by just a single lineage.

Carboniferous/Permian

Important eumalacostracan lineages appear in the fossil record through to the middle Carboniferous: the Belotelsonidea, Spelaeogriphacea and Pygocephalomorpha (Tournaisian), the Aeschronectida, Paleocaridacea and Tanaidacea (Visean), and the Cumacea (Bashkovian). Together with the appearance of the Enantiopoda (remipedes) these contribute to a mid Carboniferous peak. The extinction of the Enantiopoda and Belotelsonidea by the Kasimovian results in a decrease in observed disparity, despite the emergence of the notostracan branchiopods (Fig. 8D). There is little change in disparity throughout the Permian. A small wobble from the Ufimian to the Tatarian corresponds to the extinction of the Palaeocaridacea and the subsequent emergence of the Leptostraca.

Implied disparity (sum of ranges) reaches Recent levels by the beginning of the Carboniferous. The Eumalacostaraca are densely sampled throughout, with just two orders yet to appear. There is little change throughout the Carboniferous and Permian (or even up to the Recent).

Triassic-Tertiary

The extinction of the Pygocephalomorpha results in a barely perceptible decline in observed disparity at the start of the Triassic, followed by a larger rise in the Anisian and Carnian caused by the appearance of the anaspidaceans and the kazacharthran branchiopods. There is no subsequent change until the Jurassic (Toarcian), with the appearance of the Euzygida and the later demise of the kazacharthrans. The first records of the Caridacea and Mysidacea in the Callovian have little effect on disparity at the end of the Jurassic (Fig. 8E). A much more pronounced increase at the start of the Cretaceous corresponds to the appearance of the cladoceran branchiopods. The first fossil copepod (a parasitic siphonostomatoid) in the Aptian has little effect, as it plots within the relatively densely-occupied 'maxillipod' region of morphospace. Similar considerations apply to the first ascothoracican (cirripedes are present from the Middle Cambrian). The only remaining events are the appearance of the amphipods and anostracan branchiopods in the Priabonian, and the harpacticoid and monstrilloid copepods in the Langhian (Fig. 8F), none of which markedly influence disparity levels.

Implied disparity shows virtually no change throughout this period. The radiation of the morphologically constrained copepods (and by inference, the mystacocarids and tantulocarids) is registered, but of little overall significance.

Recent

Several morphologically rather distinct groups have no fossil record, among them the Amphionidacea, Thermosbaenacea, Bathynellacea, Mystacocarida, Tantulocarida, Branchiura, Brachypoda and Nectiopoda. These contribute to a large increase in disparity from the Pleistocene to the Recent.

The differential occupation of morphospace

Realized bodyplans

The PCO plot (Fig. 5) illustrates something that will be obvious to any carcinologist: while the crustaceans are a highly plastic group, there are large regions of morphospace that are not currently, and (from all the fossil evidence) never have been occupied. Schram (1984) demonstrated this for eumalacostracans by a simple consideration of just three parameters (the development of the cephalothoracic shield, the development of a brood pouch, and the biramy or uniramy of the thoracopods). Considering the crustaceans as a whole, no taxon has ever combined a fused cephalothorax with (for example) phyllopodous limbs or an elongate, homonomous trunk. Similarly, many specializations and unique tagmoses of the appendages occur solely within the Malacostraca. A discussion of the reasons for these limitations is beyond the scope of this study, although functional morphological and mechanical constraints may almost certainly be invoked (e.g. a remipede-like animal would be hydrodynamically inept if encumbered with a large, bivalved, cephalothoracic shield).

Morphological rarefaction

Although the distribution of all taxa (extant and fossil) in morphospace demonstrates a pattern that is clearly significantly different from random, we may still ask whether the disparity of taxa at any given time was significantly different from a similarly-sized random sample drawn from the universe of all *realized* bodyplans? In other words (for example), are the 17 orders that existed in the Moscovian more or less disparate (on average) than seventeen orders chosen randomly from the list of all the orders that existed? This question was addressed by producing rarefaction curves of all four measures of morphological variation (sums and products of ranges and variances) at sample sizes between two and one less than the total number of suborders coded. Results for sums of ranges on all coordinate axes are illustrated (Fig. 9). Values from the randomly-selected taxa are plotted as lines (mean values with upper and lower 5% confidence intervals). The disparity of the taxa actually observed at each stage is plotted as open circles and the disparity of observed plus cladistically implied taxa is plotted as triangles. Only a small number of points lie outside the 90% confidence limits. Values for observed plus implied taxa are clustered towards higher sample sizes, and are slightly closer to the mean on average.

For clarity, the difference between the average randomized values and the actual values for each stage are plotted against time (Fig. 10), with upper and lower 5% confidence intervals also indicated. Four graphs show the sum of ranges for both observed taxa (Fig. 10A,B) and implied taxa (Fig. 10C,D), on all coordinates (Fig. 10A,C) and on just the first 13 coordinates (Fig. 10B,D). All except six of the time slices (within the Cambrian, the first half of the Silurian, and the Recent), have observed values (Fig. 10A,B) exceeding the mean for samples of a similar number of taxa. A similar pattern applies to the sum of implied ranges on all coordinates (Fig. 10C). The sum of implied ranges on just the first 13 coordinates (Fig. 10D) has many more values below the mean through the Carboniferous, Permian, Cretaceous and Tertiary. Three of the graphs (B,C,D) have disparity levels below the upper 95% confidence interval, while the sum of observed ranges on all coordinates (Fig. 10A) has some values exceeding the upper interval from the Triassic to the Recent. In no cases do values fall below the lower interval (and therefore in no cases is disparity significanly less than predicted at random).

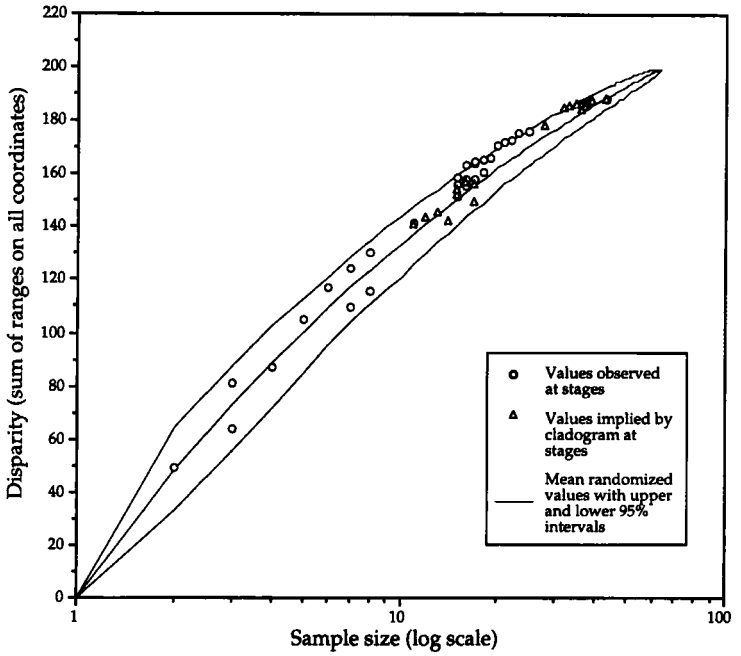


Figure 9. Rarefaction analysis of crustacean disparity. Disparity is calculated as the sum of ranges on all coordinates. Rarefaction curves have been produced from 1000 random draws (without replacement) of taxa at each sample size. The mean values, with 95% confidence intervals, are indicated as lines. Data for lineages observed at each stage are plotted as circles. Data for implied lineages (including the observed ones) are plotted as triangles. Most stratigraphic intervals contain an array of taxa that are more disparate than the average of a similar number drawn randomly from the list of all realized orders. The pattern through time is investigated in more detail in Fig. 10.

We would not expect animals to appear and disappear in morphospace purely at random. In reality, new bodyplans arise by phyletic branching, and a clade diversifies by the incremental acquisition of apomorphies along its branches. Cladograms usually reveal sequences of morphologies 'intermediate' between common ancestors and the most phenetically-derived terminals (or those supported by the largest numbers of nodes). Therefore, taxa would be expected to appear in morphospace closer (on average) to regions already occupied than predicted by a random model (provided temporal resolution was sufficient). Similarly, the early evolution of a group should be characterized by the colonization of morphospace progressively further from the common ancestor, whether in several directions or just one. Extinctions might be expected to deselect groups of related taxa, rather than picking off branches from the tree at random. Both of these effects will tend to make disparity at any given time *less* than that for a corresponding random scatter of points. Hence, the observation of greater disparity at most intervals than at random is surprising, and the fact that some of these differences are significant is even more surprising.

The Cambrian has rather lower disparity than expected at random, but not significantly so. Hence, although just a few Cambrian taxa circumscribe an appreciable fraction of Recent disparity (Fig. 7A), this is not inconsistent with the random appearance of forms. The fact that Cambrian levels are among the lowest also suggests more of a Markovian progression through morphospace than at other

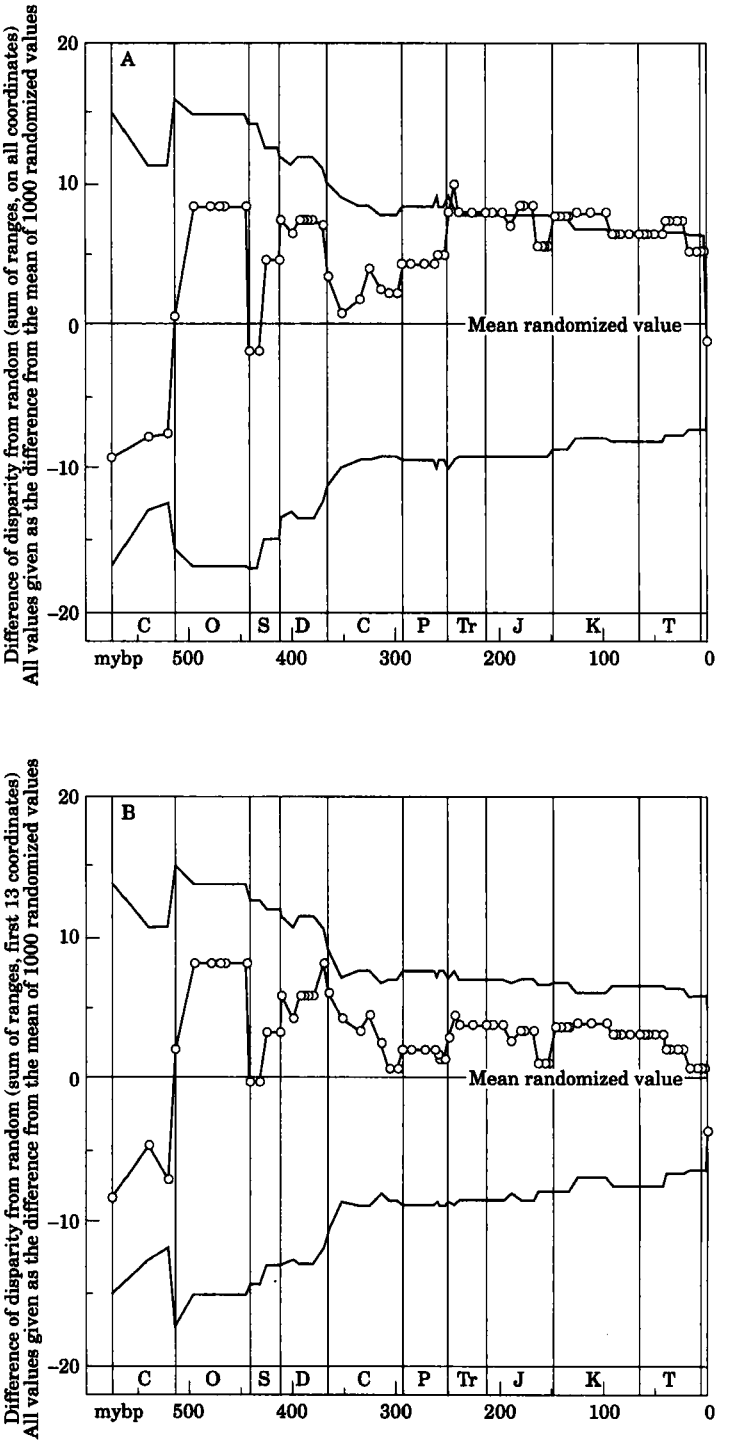
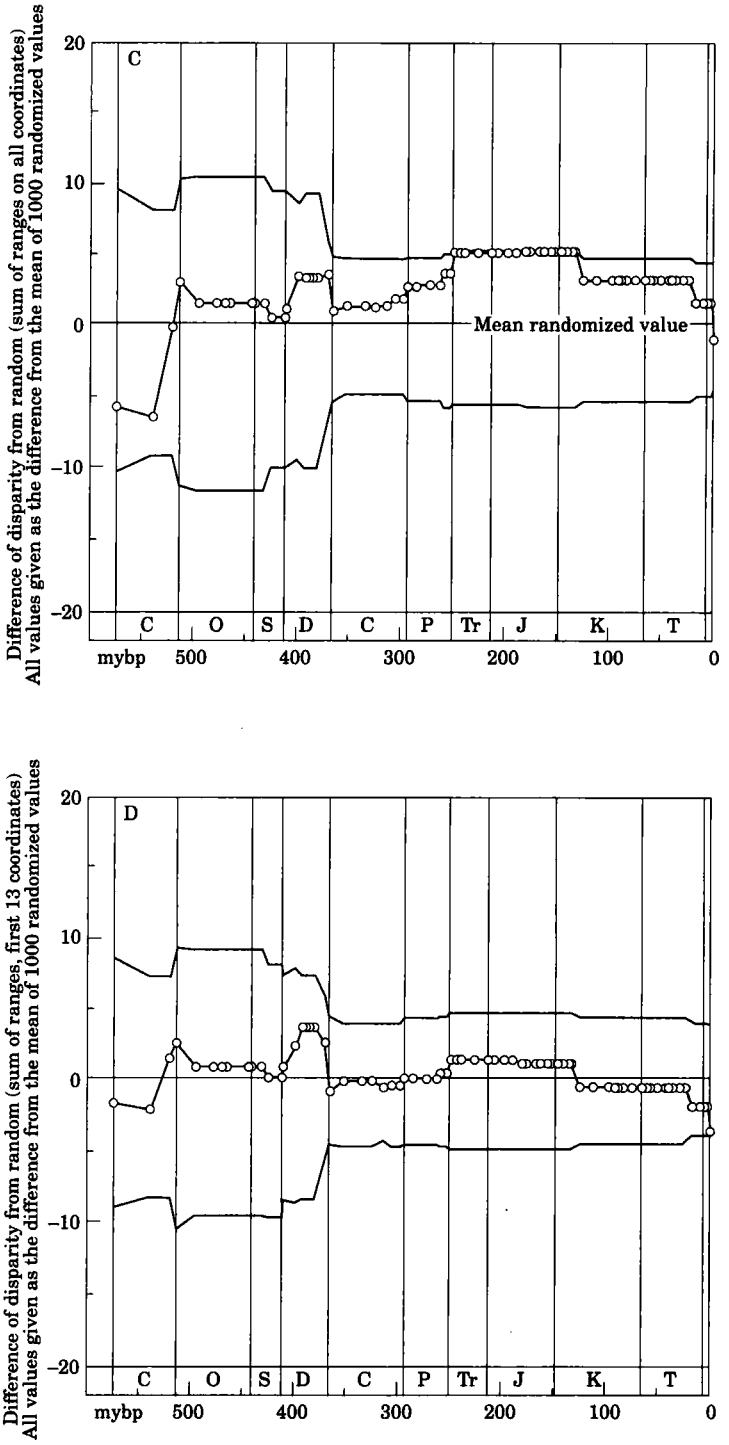


Figure 10. Deviations from randomized disparity through time. Mean randomized disparity for 1000 random draws of taxa set to zero. Significant deviations from this level ($P<5\%$) indicated as unbroken lines. Actual deviations indicated by open circles. Disparity of observed lineages calculated as (A,C) the sum of ranges on all coordinates; (B,D) the 13th root of the product of ranges of the first 13 coordinates. See text for discussion.



times. These observations, though dampening the extreme 'explosive' model (Gould, 1989, 1991), still attest to a markedly scattered initial record for the Crustacea, and some implications of this are discussed later.

An Ordovician peak in observed levels (Fig. 10A,B) contrasts with near-random implied levels (Fig. 10C,D). Numbers of observed taxa through the Ordovician are low, but many of the cladistically-implied lineages are close to the morphology of observed ones. Another peak in the Devonian (greatest for the implied differences, Fig. 10C,D) marks the radiation of the branchiopods and cephalocarids. The appearance of the highly-derived reptantian eumalacostracans in the record at the end of the Devonian sees the highest overall observed level on the first 13 coordinates (Fig. 10B) (and the one closest to being significant). The cladogram implies the appearance of numerous other (morphologically relatively similar) eumalacostracan lineages at this time, which reduces the difference from random in implied disparity (Fig. 10C,D). Implied disparity shows a gradual increase through the Carboniferous and Permian, and in Fig. 10C (all coordinates) reaches levels on the border of significance through the Triassic, Jurassic and early Cretaceous. Most major clades contain many modern orders with lineages or ghost lineages extending back to the Triassic, with the exception of the maxillopods. The implied radiation of this latter group (including the morphologically rather conservative copepods, Fig. 6) in the Aptian sees implied disparity levels decrease closer to the random level. Towards the Recent, the appearance of several ghost lineages (and many observed lineages) close to regions of morphospace already occupied sees a decrease in disparity to levels well below the mean average.

In general, the graphs based on just the first 13 coordinates (Fig. 10B,D) show less deviation from random than the corresponding graphs based on all positive coordinates (Fig. 10A,C). This suggests that the spread of taxa on many of the later coordinates is more eccentric than on the earlier ones. While the first few coordinates appear to differentiate between the major clades, later coordinates probably pull out differences within these groups (lower portions of the overall variance). This in turn suggests that the radiation *within* crustacean clades may differ more from random than the radiation of the clades themselves. It is also noteworthy that the pattern of observed disparity calculated on all coordinates differs most from that calculated on just the first 13 coordinates in the period Triassic to Recent (where all coordinates imply greater and often significant deviation from random). This interval sees the appearance of many important new morphologies within the Eumalacostraca and Maxillopoda. The corresponding period of greatest difference for implied (closer to the truth?) disparity lies from the Carboniferous to the Recent, and is undoubtedly driven initially by the radiation of the Eumalacostraca.

'Mass extinctions', diversity, disparity and sampling

With the exception of the event at the end of the Cambrian, levels of disparity are undiminished at times with significantly-elevated extinction rates. The end-Permian, end-Triassic and end-Cretaceous events cause no changes in the sums and products of ranges, while the end Ordovician sees an increase. Disparity increases dramatically from the late Devonian to the early Carboniferous.

The differential effects of mass extinctions on 'diversity' at various levels in the taxonomic hierarchy are well-documented (Raup & Sepkoski, 1986; see also Smith & Jeffery, 1998, for a consideration of some ecological factors influencing clade survivorship). Species and genera may be wiped out with relative ease, but extinguishing a family means knocking out perhaps tens or even hundreds of genera, and the demise

of an order implies the loss of many families. In general, as we increase the hierarchical level of the taxa we study, perturbations in taxic diversity decrease.

Disparity (or bodyplan variety) is undoubtedly more closely tied to numbers of higher taxa (orders, classes etc.) than to numbers of species (Gould, 1989), but there are serious dangers in simply equating the two (Briggs *et al.*, 1992). Modern coelacanths and xiphosurans, for example, are lone representatives of what were once vastly more diverse higher taxa *and* bodyplan designs. In effect, just a few genera 'hold the fort', bolstering the disparity of the larger groups that contain them. To this extent, the demise of either of these 'living fossils' would be of greater significance than the extinction of (for example), a genus of gobies or three genera of spiders. Similarly, some crustacean orders are monogeneric (e.g. Amphionidacea, Procarididea) while others contain hundreds or thousands of genera (e.g. Isopoda, Reptantia). This is why variance-based indices of morphological variation are considered less informative for the present purposes; rare and eccentric bodyplans can be lost with very little change in the indices, and the absolute magnitude of change will depend on the extent of taxonomic splitting.

A worthwhile exercise (and one, no doubt, destined for further study) would be to investigate crustacean disparity by sampling below the ordinal and sub-ordinal levels. It is quite possible, however, that this would yield patterns of range sums and products broadly similar to those outlined here. Some regions of the morphospace would be much more densely occupied than they are in the present analysis (which might have some implications for the orientation of the cloud of points along the coordinate axes, and variances along those axes). However, in terms of the character set used here, further taxonomic splitting would reveal few additional combinations of states, and therefore do little to change the overall distribution of occupied regions within the space. Sums and products of ranges (the preferred indices of disparity) would be expected to change little, while sums and products of variances would be more influenced by the differential clustering of points (Foote, 1992b).

Morphological variety profiles in trilobites

The most prominent examples of studies similar to this are found within a series of papers on trilobites by Foote (1989, 1990, 1991a, 1992b, 1993). The trilobites are markedly less disparate overall than the crustaceans (Wills *et al.*, 1998a), and are a clade of much lower rank [the Crustacea oppose the Arachnomorpha, and this latter clade contains not only the trilobites but also the chelicerates and many other Burgess 'problematica' (Wills *et al.*, 1998b)]. Species and specimens were selected in a rather more 'haphazard' fashion than here. Most of Foote's investigations focussed on the average dissimilarity between forms (variance) rather than the range of morphospace occupied, and in some cases, 'outliers' were arbitrarily ignored (Foote, 1991a,b). A direct comparison with disparity (as defined here for the Crustacea) is not straightforward. Moreover, the data used by Foote differed quantitatively and qualitatively from those employed here. Differences in the morphology of the cranidium alone were used to define a trilobite morphospace, rather than attempting to capture morphological variation in the whole body. Instead of a suite of discrete morphological characters, cranidial outlines were projected into two dimensions, and described by a series of harmonic coefficients from a Fourier transform (Foote, 1989). Shape variation in a single, conservative feature was then used as a proxy for differences in overall

bodyplan morphology [a big assumption, but one which probably has some basis (Foote, 1989, 1991a)]. Despite the differences in approach, some interesting observations arise from a comparison of patterns of morphological variety in the two studies.

The sum of variances on all axes for all trilobites was about one third of its maximum in the first Cambrian time-slice (Foote, 1993). Visual inspection of the ordinations suggests that the sums of ranges (at least on the first two principal components) may have been in a similar ratio. Trilobites, like crustaceans, appear to have been morphologically rather diverse early in their recorded history. The variance sum is seen to decrease later in the Cambrian, but reached its all-time maximum in the latter half of the Ordovician. Illustrations suggest that the sum of ranges probably peaked at the same time. Sums of variances for crustaceans were also at higher levels through the Ordovician. The early histories for the two groups therefore appear to be quite similar. Thereafter, however, the patterns differ markedly.

Trilobites maintained a fairly dispersed but much more rarefied distribution through the Silurian and Devonian, but sums of variances (and sums of ranges) significantly decreased. By contrast, crustaceans showed an increase to an all time-high on all indices by the end of the Devonian (except the sum of variances, which peaked in the early and mid Devonian). The early Carboniferous saw a marked crash in variance sums for trilobites, and by the end of the period the group occupied a very restricted pocket of morphospace (certainly a smaller sum of ranges than in the Cambrian). The last trilobites died out at the end of the Permian. Disparity (the sum of ranges) for crustaceans continued to increase steeply until the Moscovian (86% of the Recent), with just a slight overall upward trend from the end of the Carboniferous to the end of the Tertiary. [The filling in of morphological intermediates during the Carboniferous caused the sum of variances to decrease to a level around that in the Ordovician (where it stayed until the Recent)].

Hence, the Silurian to Carboniferous demise of trilobite disparity coincided with a major phase of disparity increase for the Crustacea. It is tempting to speculate that the phenomena may be linked. Ecological competition is difficult to demonstrate in the living fauna, and much more so in the fossil record (Benton, 1996). Trilobites, like crustaceans, exploited a variety of habitats and niches. However, the majority were benthic, either crawling over the sediment or burrowing through surface layers. Most of these were probably scavengers or direct deposit feeders, while a few may have been active predators, ambushing prey from a partially buried hiding place (Bergström, 1973). Some genera may also have suspension fed. Most benthic trilobites were a few centimeters long, although some reached 60 cm or more. A few trilobites under 1cm were planktonic. Crustaceans exploited most of the same habitats even in the Cambrian, but benthic scavengers, deposit feeders and predators seem only to have taken off with the radiation of the Eumalacostraca in the late Devonian and Carboniferous (corresponding with the rapid decline in trilobite disparity). The first eumalacostracans were almost certainly active walkers rather than swimmers (Manton, 1953; Dahl, 1992), and the differentiation of the thorax and abdomen is likely to have been associated with an epibenthic mode of life (Calman, 1909).

It may also be significant that the diversity of eurypterid ('water-scorpion') families increased from the mid Ordovician to a Silurian peak, and then declined markedly in the Devonian (Selden, 1993). Eurypterids appear to have used a variety of feeding strategies, although most were rapacious carnivores. One might speculate tentatively on the possibility of competition between eurypterids and the reptant decapods and hoplocarids.

Briggs, Fortey & Clarkson (1988) noted some complementarity in the generic diversity curves for Eurypterids and fish, with fish diversity peaking in the late Devonian. The rise of gnathostomes (the fish group most 'adaptively comparable' with the predaceous eurypterids) coincided with the onset of eurypterid decline in the Silurian-early Devonian. However, the decline in eurypterids was not complemented by an increase in fishes, but rather paralleled by a similar decline (Briggs, 1985). Briggs *et al.* (1988) concluded that there is little evidence to support the hypothesis of competitive interaction between these two groups, and cautioned against an uncritical reading of the fossil record in this way.

Implications for the Cambrian 'explosion'

Even though disparity of the earliest Cambrian crustacean fossils is indistinguishable from a similarly-sized random sample (Fig. 10), this is remarkable. Even if functional and developmental constraints took effect early in the history of a group, initial diversification should refelct a progressive exploration of morphospace, rather than an apparently simultaneous colonization of several extremes. Such drift could originate anywhere within the space, centrally, peripherally (the remipedes, and all other groups considered plesiomorphic, plot in the top left of Fig. 5), or somewhere in between. Since the earliest fossils are as disparate (on average) as a similar number of ordinal-level bauplane picked randomly from the list of all those realized (i.e. Lower Cambrian fossils are widely scattered in morphospace), this strongly suggests that the earliest diversification of the group is not recorded here. This may be either because the event was explosive (and the fossil record samples the result on too coarse a time interval to pick it up), or because significant cladogenesis took place in the Precambrian (although this is not evident in the fossil record). Fortey, Briggs & Wills (1995, 1997) discussed the latter possibility for all arthropods, and all Metazoa, proposing a Precambrian history of cladogenesis in animals too small to be fossilized. As discussed above, this would require the roughly simultaneous triggering of size increase in many lineages by some (presumably external) factor. Support for this idea comes from the observation that many phyla have minute (<1 mm) sister groups or basal representatives. Recent rate studies in several molecules (haemoglobin, NADH dehydrogenase subunit 1, 18S rRNA, ATPase 6, cytochrome c and cytochrome c oxidase subunits I and II) imply divergence times for all phyla in excess of 1200 million years before present (Wray *et al.*, 1996). A more conservative molecular estimate (Ayala, Rzhetsky & Ayala, 1998) dates the protostome/deuterostome split to about 670 million years, still implying a Precambrian history for modern animal groups of 120 million years or more. The *relatively* low levels of disparity through the Cambrian are, perhaps, even more surprising given claims about the inordinately disparate morphologies of animals from the Burgess Shale, Chengjiang and Peary Land faunas (Gould, 1989, 1991).

CONCLUSIONS

(1) Historically, palaeontologists have utilized two sources of data for inferring phylogeny; morphological and stratigraphic. Stratigraphic data have usually been ignored by most cladists, who deduce phylogeny from the distribution of

character states alone. Recently a variety of techniques have been devised that utilize stratigraphic congruence within the optimality criteria for cladogram selection. Alternatively, morphological and stratigraphic information can be considered independently, allowing robust phylogenies to identify ghost ranges and gaps in the fossil record. Randomization procedures offer a significance test of the 'goodness of fit' of these two types of data.

(2) Cladistic and appropriate morphospace methods can be applied in parallel to data sets of discrete morphological characters.

(3) The cladogram for crustaceans does not conform precisely to any previously published tree for the clade (Schram, 1986; Brusca & Brusca, 1990). Malacostracans (*sans* phyllocarids) constitute a sister group to maxillopods. Phyllopod taxa resolve as a paraphyletic series of sister groups in opposition to this clade (interspersed with 'Orsten' genera). The putatively ancestral forms, branchiopods (Fryer, 1992) and cephalocarids (Hessler & Newman, 1975) fall low in the tree. The hoplocarids are probably not the sister group of the eumalacostracans (Schram, 1969), but are most parsimoniously interpreted in association with the eucarids. The peracarids form a clade, but the syncarids are a paraphyletic taxon, united by symplesiomorphic features. The cladogram implies the existence of significantly shorter ghost ranges than obtained from a random permutation of stratigraphic data over the entire tree. However, stratigraphic congruence within the Eumalacostraca is significantly worse than random using both the tree presented here, and published alternatives. One of several possible explanations for this is the absence of fossils because numbers of individuals have always been small (living examples of several lineages have only been discovered relatively recently).

(4) Three broad 'clusters' emerge from the morphospace plot, echoing the structure of the cladogram. The first comprises the maxillopods and 'Orsten' taxa (with the exception of *Rehbachiella*). The second contains the malacostracans, leptostracans and archaeostracans. The third encompasses several taxa believed to be close to the ancestral crustracean morphology (Remipedia, Cephalocarida, Branchiopoda) plus the Canadaspidida, *Odaraia* and *Waptia*. Remipedes (used to root the cladogram) lie at one extreme of the MST, and in a far corner of the plot. There must have been a strongly directional component to the early exploration of morphospace, irrespective of precisely where the ancestor lay. Malacostracans extend along the second coordinate, from the more entomostracan-like syncarids and thermosbaenaceans, to the typically caridoid reptantians, eukyphids, and dendrobranchiates at the tips of the MST. Several regions of the overall morphospace remain conspicuously unoccupied. However, there are few obvious discontinuities in those regions that *are* occupied (i.e. there are many intermediate forms).

(5) The maxillopods and eumalacostracans show a comparable disparity (sum of ranges) overall, while the phyllopods (paraphyletic) are slightly more disparate. The branchiopods are almost as disparate as the maxillopods, although only five branchiopod exemplars were coded, compared with 21 maxillopods. Correcting for sample size differences, the branchiopods, maxillopods and eumalacostracans all have a comparable disparity.

(6) In the Lower Cambrian, the levels of disparity for observed fossil lineages were about a third of their present value (using range-based measures). Observed disparity increased throughout the Cambrian but after a marked slump in the Tremadocian, stabilized to a level similar to the Lower Cambrian thoughout the rest of the Ordovician. Increases through the Silurian register the appearance

of ostracods and the anostracans. The Devonian witnessed two marked increases with the diversification of branchiopods and eumalacostracans. Disparity attained about 85% of Recent levels by the base of the Permian, and showed a gradual increase to 94% by the Pleistocene. Values incorporating ghost lineages exhibit higher initial levels, with less marked peaks and troughs and fewer perturbations overall. Levels of disparity are not significantly diminished at most times of mass extinction (with the exception of the end Cambrian). Most time intervals contain an assemblage of orders with a disparity exceeding the mean for a similarly-sized random sample. However, this difference is only significant in the absence of ghost ranges, intermittently from the Triassic to the mid Tertiary.

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